

A Slow Burn: The Opioid Epidemic and the Rise in Teen Fertility

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Abstract

In this article, we establish a causal connection between the U.S. opioid epidemic and teen birth rates—a relationship that the existing literature on opioids and fertility has left unexplored. Drawing on the identification strategy of [Arteaga and Barone \(2025\)](#), we exploit the fact that OxyContin was initially approved and marketed for cancer pain in 1996, so that areas with higher pre-1996 cancer mortality rates received disproportionately greater opioid exposure. Using an annual panel of 621 commuting zones over 1990–2020 with commuting zone and state-by-year fixed effects, we find a delayed and escalating effect on teen fertility. Teen birth rates in more exposed areas show no differential trend through 2000, then increase by 2.5 births per 1,000 women aged 15–19 during 2001–2004, peak at 4.2 per 1,000 during 2005–2007, and remain elevated at 3.5 per 1,000 during 2008–2014. The peak effect represents a 10% increase relative to the sample mean. Heterogeneous effects are largest in commuting zones with high Black population shares and concentrated in the South and West. Complementary stacked first-difference estimates using Census/ACS data confirm that opioid exposure increased general fertility. We investigate ten potential mechanism channels and find that seven produce results consistent with theoretical predictions. The delayed dynamic of these results points to slow-acting community-level mechanisms including intergenerational transmission of disadvantage, cumulative erosion of parental supervision, and deepening economic despair that reduced the perceived opportunity cost of early childbearing.

JEL Codes: I12, I18, J13, J12

Keywords: Opioid crisis, teen pregnancy, fertility, OxyContin, commuting zones, mechanisms

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1 Introduction

The opioid epidemic stands as one of the most devastating public health crises in modern American history. Between 1999 and 2021, more than 500,000 Americans died from opioid-involved overdoses, a toll that exceeds the combat deaths of every major American war since Vietnam combined (Maclean et al., 2022). The crisis has touched nearly every dimension of American life, and a growing body of rigorous economic research has documented its far-reaching consequences. Case and Deaton (2015, 2017) demonstrated that rising “deaths of despair”—driven substantially by opioid overdoses—reversed decades of mortality improvement among middle-aged White Americans without a college degree. Krueger (2017) linked the opioid epidemic to the secular decline in prime-age male labor force participation, a finding elaborated by Aliprantis et al. (2023), who estimate that opioids account for 44 percent of the decline in men’s labor force participation between 1999 and 2015. The crisis has spilled into child welfare systems (Buckles et al., 2023), increased criminal activity and incarceration (Sim, 2023), strained municipal finances (Cornaglia et al., 2022), and contributed to political realignment in affected communities (Arteaga and Barone, 2025). Yet amid this extensive documentation of the opioid epidemic’s consequences, a critical gap remains: no study has causally estimated the effect of the opioid crisis on teen fertility. This is a remarkable omission. Teen childbearing carries well-documented costs for mothers and children alike—lower educational attainment, reduced lifetime earnings, and worse developmental outcomes for children (Hotz et al., 2005; Kearney, 2023)—and multiple theoretical channels predict that a sustained community-level health crisis like the opioid epidemic should increase teen birth rates. This article fills that gap.

The absence of causal evidence on teen fertility is all the more puzzling given that closely related work has examined the broader fertility effects of opioids. In a recent working paper, Arteaga et al. (2025) estimate that opioid exposure increased general fertility rates by approximately 10 percent, a large and statistically significant effect. However, when those authors examined age-specific fertility, they found no significant effect among teenagers, leading them to conclude that the general fertility increase was concentrated among women in their twenties and thirties. Simi-

larly, [Caudillo and Villarreal \(2021\)](#), in a sociological study of the opioid epidemic’s demographic consequences, noted that adolescents appeared to be the “least affected” age group. [Ko and Powell \(2025\)](#) find that opioid-induced fertility actually *declined* among reproductive-age women, a result driven by the direct pharmacological effects of opioid use on fecundity and sexual function. The only evidence linking opioids specifically to teen births is correlational: [Durr et al. \(2021\)](#) document a positive association between opioid prescribing rates and teen pregnancy rates across 55 counties in West Virginia, but their analysis cannot distinguish causation from the confounding forces that simultaneously drive both opioid use and teen fertility in economically distressed Appalachian communities. The central reason that the teen fertility effect has eluded prior causal research, we argue, is that the effect emerges with a five-to-ten-year lag that is invisible in shorter panels, pooled difference-in-differences specifications, or analyses that restrict the post-treatment window to a narrow band of years around the initial OxyContin introduction.

The main results reveal a delayed, escalating, and eventually attenuating effect of opioid exposure on teen fertility. The event study estimates show no differential pre-trend: the joint F-test for the pre-treatment coefficients (1990–1996) yields $F = 1.17$ with $p = .319$, confirming that high- and low-cancer-mortality commuting zones were on parallel teen birth trajectories before OxyContin’s introduction. The year-by-year coefficients remain statistically indistinguishable from zero through 2000, consistent with the hypothesis that the initial wave of opioid prescribing did not immediately affect teen fertility. Beginning in 2001, the estimates turn positive and grow steadily. Grouping the post-treatment period into theoretically motivated intervals, we find that teen birth rates in more-exposed commuting zones increased by 2.5 births per 1,000 women aged 15–19 during 2001–2004 (significant at the 10 percent level), peaked at 4.2 per 1,000 during 2005–2007 (significant at the 1 percent level), and remained elevated at 3.5 per 1,000 during 2008–2014 (significant at the 5 percent level). The peak grouped-period estimate of 4.2 per 1,000 represents approximately 10 percent of the sample mean teen birth rate of 40.7, a substantial and economically meaningful magnitude. After 2014, the effect attenuates to 1.5 per 1,000 and loses statistical significance, potentially reflecting the nationwide intensification of teen pregnancy prevention ef-

forts, the spread of long-acting reversible contraceptives (LARCs), and the compositional shift in the opioid crisis toward illicit fentanyl, which kills users rather than sustaining them in communities where they erode family and social structures.

The delayed dynamic pattern is the intellectual core of this article, and it distinguishes the opioid channel from other contemporaneous shocks that have been studied in the fertility literature. When [Autor et al. \(2019\)](#) examine the effects of the China trade shock on marriage and fertility, they find immediate marriage market responses: import competition reduces male employment, lowers marriage rates, and increases the share of children living in single-parent households within the same decade. The opioid-teen-birth relationship, by contrast, exhibits a five-to-ten-year lag between the initial exposure and the emergence of significant fertility effects. This delayed pattern is consistent with a set of slow-acting, community-level mechanisms that require time to accumulate. First, parental addiction erodes household supervision gradually: a parent who begins misusing OxyContin in 1998 may not reach the point of incapacitation or removal from the household for several years, and the children most affected are those who pass through adolescence during the period of peak family dysfunction. Second, schools and public services deteriorate progressively as opioid-affected communities lose tax revenue, face rising demand for social services, and experience teacher and counselor burnout. Third, the intergenerational channel operates on a biological clock: children born to or raised by opioid-using parents reach their own teen years roughly 13 to 18 years after their parents' initial exposure. Fourth, social norms around early childbearing may shift slowly as the cumulative weight of community distress reduces the perceived opportunity cost of teen motherhood ([Case and Deaton, 2015](#); [Kearney and Levine, 2012](#)). The finding by [Arteaga et al. \(2025\)](#) that opioids had no significant effect on teen births is fully reconcilable with our results: their analysis relies on a shorter post-treatment window and does not disaggregate the post-period into the fine-grained intervals that reveal the delayed escalation. Age-disaggregated data and a longer panel are essential for detecting this effect.

The effects are heterogeneous across racial composition and geography. We split the sample of commuting zones by their pre-period population shares of White, Black, and Hispanic residents

and estimate the grouped-period specification separately for each tercile. Commuting zones in the top tercile of Black population share exhibit the largest peak effect, reaching 6.3 additional teen births per 1,000 during 2005–2007 (significant at the 1 percent level), consistent with the well-documented racial disparities in both opioid exposure and teen birth rates. Commuting zones with high White or Hispanic population shares show more persistent effects that extend into the 2015–2020 period, possibly reflecting the geographic spread of the opioid crisis into rural White communities during the second and third waves of the epidemic. Geographically, the effects are concentrated in the South and West, where prescription opioid use was highest and public health infrastructure was weakest. The Northeast, by contrast, shows negative point estimates throughout the post-period, a pattern that may reflect stronger public health systems, earlier adoption of naloxone distribution programs, and more robust family planning infrastructure in that region.

To complement the annual event study, we estimate stacked first-difference models using decennial Census and American Community Survey data following the framework of [Autor et al. \(2013\)](#). These models examine general fertility outcomes for women aged 20–39 across 722 commuting zones over two stacked periods (1990–2000 and 2000–2014). The results confirm that opioid exposure increased general fertility: the birth rate among women aged 20–39 rose by 0.79 births per 1,000 (significant at the 10 percent level), and the share of women with children in the household increased by 0.30 percentage points (significant at the 5 percent level). The share of mothers who were unmarried increased by 0.10 percentage points but was not statistically significant. These results are broadly consistent with the findings of [Arteaga et al. \(2025\)](#) and serve as a validation exercise confirming that the opioid exposure measure captures meaningful variation in fertility behavior in an independent dataset. Importantly, these results are robust to controlling for the China trade shock, indicating that the opioid and trade channels operate independently.

We investigate ten potential mechanism channels through which the opioid epidemic may have increased teen fertility. For each channel, we identify a theoretical hypothesis, select the best available proxy variable, specify the expected sign and timing of the effect, and estimate either a grouped-period event study (for annual data) or a stacked first-difference model (for Census/ACS

data). The ten channels span proximate behavioral pathways (parental supervision, family disruption, foster care displacement, teen substance use), structural economic forces (missing men through incarceration and mortality, economic despair and reduced opportunity cost, educational disruption), public service deterioration (diversion of family planning resources), psychological deterioration (mental health and ACE accumulation), and social amplification (peer effects and contagion). Of the ten channels, seven produce results that are consistent with the theoretical predictions in at least one key outcome variable and during the relevant post-treatment period. The strongest results emerge for incarceration (jail population surges by 173 per 100,000 after 2015), single-parent household formation (+0.39 percentage points, significant at the 1 percent level), unemployment (+0.85 percentage points, significant at the 1 percent level), and labor force participation decline (-0.8 percentage points, significant at the 1 percent level). The consistent timing across these diverse mechanisms—all showing their strongest effects during the same 2005–2014 window when teen birth effects peak—provides the most compelling evidence for the slow-acting, multi-channel interpretation of the main results.

The remainder of this article proceeds as follows. Section 2 provides background on the OxyContin channel and the theoretical mechanisms linking opioids to teen fertility. Section 3 describes the data sources and construction of the analysis sample. Section 4 presents the empirical strategy. Section 5 reports the main results, including heterogeneity analyses. Section 6 investigates ten potential mechanism channels. Section 7 presents robustness checks. Section 8 concludes.

2 Background and Motivation

2.1 The OxyContin Channel

Crucially for the identification strategy, OxyContin was initially approved and marketed specifically for the management of moderate-to-severe pain associated with cancer. The FDA’s initial labeling restricted the drug’s indicated use to chronic pain conditions that required continuous, around-the-clock opioid treatment, and Purdue’s early marketing efforts were concentrated on on-

cologists, palliative care specialists, and pain management clinics serving cancer patients. Purdue deployed an extensive network of pharmaceutical sales representatives—its sales force nearly doubled between 1996 and 2000—who targeted physicians in areas with the highest concentrations of cancer patients. The company sponsored pain management conferences, distributed branded promotional materials, and offered starter coupons for free prescriptions, all with a focus on the cancer pain market ([Arteaga and Barone, 2025](#)). This targeting strategy means that the geographic distribution of initial OxyContin prescribing was heavily influenced by the pre-existing distribution of cancer mortality across American communities.

The resulting geographic variation in opioid exposure provides the basis for identification. Areas with higher cancer mortality rates in the years immediately preceding OxyContin’s introduction—measured here using 1995 age-adjusted cancer mortality rates from the CDC—received disproportionately more OxyContin prescriptions during the initial marketing push. As the drug diffused beyond its original cancer-pain indication and into the broader chronic pain market during the late 1990s and early 2000s, these initially more-exposed communities maintained their higher levels of opioid prescribing. The cancer mortality rate thus serves as a continuous treatment intensity measure, capturing the degree to which a community was exposed to the first wave of the opioid epidemic. The plausibility of this instrument rests on the assumption that, conditional on the controls included in the empirical specification (commuting zone fixed effects and state-by-year fixed effects), the pre-1996 cancer mortality rate is uncorrelated with unobserved determinants of teen fertility trends. [Section 4](#) discusses this identifying assumption in detail and presents evidence supporting its validity.

The opioid epidemic unfolded in three distinct waves, each with different geographic and demographic characteristics, and this three-wave structure is essential for interpreting the dynamic pattern of teen birth effects. The first wave (1996–2010) was driven by prescription opioids, primarily OxyContin and other pharmaceutical products. National opioid prescribing rates rose steadily from the late 1990s, peaking in 2012 at 81.3 prescriptions per 100 persons ([Maclean et al., 2022](#)). This wave was concentrated in communities with high cancer burdens, high rates of

occupational injury (coal mining, manufacturing), and aggressive pharmaceutical marketing. The second wave began around 2010, when Purdue reformulated OxyContin to make it more difficult to crush and snort, and the FDA imposed tighter restrictions on opioid prescribing. Many individuals who had become dependent on prescription opioids transitioned to heroin, which was cheaper and more readily available (Evans et al., 2019; Alpert et al., 2018). Heroin overdose deaths tripled between 2010 and 2015. The third wave, beginning around 2013, was driven by the introduction of illicitly manufactured fentanyl and its analogs into the drug supply. Fentanyl is 50 to 100 times more potent than morphine, and its appearance caused a dramatic acceleration in overdose deaths (Maclean et al., 2022).

This three-wave structure has direct implications for the interpretation of teen birth effects. During the first wave, prescription opioid users remained largely functional—many held jobs, maintained households, and kept children in school, even as their dependence deepened. The community-level consequences of widespread opioid use—family dissolution, economic decline, institutional deterioration—accumulated gradually. It is during this slow accumulation phase that the conditions conducive to elevated teen fertility were being established. By the time the second and third waves arrived, bringing higher mortality and more visible social disruption, the teen birth effect was already at or past its peak. The attenuation of the teen birth effect after 2014 may reflect both the lethal nature of the later waves (which removed opioid users from communities through death rather than sustaining them in states of chronic dysfunction) and the nationwide intensification of teen pregnancy prevention efforts, including expanded access to long-acting reversible contraceptives.

2.2 Theoretical Mechanisms Linking Opioids to Teen Fertility

The theoretical literature on teen fertility points to several channels through which a sustained community-level health crisis like the opioid epidemic could increase teen birth rates. We organize these into four broad categories, each of which generates testable predictions about the timing and magnitude of the expected effects.

Parental supervision and family disruption. The most proximate channel runs through the household. When a parent becomes addicted to opioids, the quality of parental supervision declines: the addicted parent is less able to monitor an adolescent’s activities, enforce curfews, communicate about sexual health, and provide the emotional stability that delays early sexual activity. The empirical literature consistently finds that parental monitoring is one of the strongest protective factors against teen pregnancy ([Kearney and Levine, 2012](#)). When addiction progresses to the point of parental incapacitation, incarceration, or death, the adolescent may be placed with relatives, in foster care, or left in a household headed by a remaining parent who is overwhelmed by the demands of single parenthood. Youth in foster care face dramatically elevated rates of teen pregnancy—[Doyle \(2007, 2008\)](#) documents the causal effects of foster care placement on a range of adverse outcomes including teen motherhood. The foster care system itself came under enormous strain during the opioid crisis: entries into foster care increased by 10 percent nationally between 2012 and 2017, with increases exceeding 30 percent in the hardest-hit states ([Buckles et al., 2023](#)). The key prediction of this channel is that the teen birth effect should be delayed: children must first be exposed to parental opioid use, the addiction must progress to a severity that meaningfully impairs parenting, and the affected children must then reach adolescence. This timeline is broadly consistent with a 5–10 year lag.

Missing men and marriage market thinning. The opioid epidemic has removed large numbers of young men from communities through incarceration, disability, and death. Drug-related arrests and incarceration surged as the epidemic intensified, particularly after the transition to heroin and fentanyl triggered aggressive law enforcement responses. [Charles and Luoh \(2010\)](#) show that male incarceration reduces the supply of marriageable men, lowering marriage rates and increasing out-of-wedlock fertility. [Wilson \(1987\)](#) first articulated the “marriageable men” hypothesis in the context of deindustrialization: when the pool of employed, non-incarcerated men shrinks, women face diminished returns to delaying childbearing until marriage, and non-marital fertility rises. The opioid epidemic may have generated a similar dynamic, particularly in communities where

young men were disproportionately affected by addiction, incarceration, and overdose death. To the extent that teen girls in these communities perceived fewer viable future marriage partners, the opportunity cost of early, non-marital childbearing would have fallen.

Economic despair and reduced opportunity cost. The most comprehensive theoretical framework for understanding variation in teen birth rates is the opportunity cost model developed by [Kearney and Levine \(2012, 2014\)](#). In this framework, teen pregnancy rates are highest in communities where young women perceive the least gap between their expected life trajectory as a teen mother and their expected trajectory if they delay childbearing. When economic prospects are dim—when unemployment is high, wages are low, educational institutions are weak, and upward mobility is rare—the perceived cost of early motherhood falls, and teen birth rates rise. The opioid epidemic created precisely these conditions. Communities with high opioid exposure experienced declining labor force participation ([Aliprantis et al., 2023](#)), rising unemployment, increasing dependence on public assistance, and a pervasive sense of economic and social decline that [Case and Deaton \(2015\)](#) capture under the rubric of “despair.” This channel predicts that teen birth effects should intensify as the cumulative economic consequences of the opioid crisis deepen over time.

Deterioration of public services and family planning access. The opioid crisis strained the budgets and capacity of public institutions, including the health departments and family planning clinics that provide contraceptive services to low-income women. [Packham \(2017\)](#) demonstrates that closures of publicly funded family planning clinics increase teen birth rates by 1.3 percent per clinic lost, and [Kearney and Levine \(2009\)](#) show that Medicaid expansions of family planning services reduce teen births significantly. As local and state governments redirected resources toward addiction treatment, naloxone distribution, and the cascading demands of the child welfare system, family planning services may have received less funding and attention, reducing contraceptive access for the very populations most vulnerable to elevated teen fertility.

A unifying feature of these four theoretical channels is that each predicts a delayed effect—an interval of years between the initial increase in opioid prescribing and the emergence of measurable

impacts on teen birth rates. This prediction distinguishes the opioid channel from contemporaneous shocks (such as trade liberalization or recessions) that typically generate more immediate fertility responses. The delayed pattern therefore serves as a testable implication that, if confirmed by the data, provides additional support for the causal interpretation of the estimates. As we show in Section 5, the data strongly confirm this prediction: the teen birth effect emerges roughly five years after OxyContin’s introduction, peaks after nine to eleven years, and persists for nearly two decades.

3 Data

3.1 Teen Birth Rates (1990–2020)

The primary outcome variable is the teen birth rate, defined as the number of live births to women aged 15–19 per 1,000 women in that age group. Constructing this variable at the commuting zone level over a thirty-year period requires combining two data sources that use different methodologies, and we describe each in turn.

For the period 1990–2002, we construct teen birth rates from the NBER Natality Detail Files, which contain individual-level records of every birth registered in the United States. Each record includes the mother’s age, county of residence, and demographic characteristics. We aggregate births to mothers aged 15–19 by county, map counties to commuting zones using the crosswalk of [Autor et al. \(2013\)](#), and divide by the female population aged 15–19 in each commuting zone (obtained from intercensal population estimates) to obtain rates per 1,000. This construction follows standard practice in the teen fertility literature and has the advantage of being based on a complete census of births rather than a sample.¹

The splice between the two data sources occurs at the 2002–2003 boundary. To validate the continuity of the combined series, we compare the NBER-constructed and NCHS-smoothed rates

¹We exclude births with missing maternal age (approximately 0.1 percent of records) and births with missing county codes (approximately 1.5 percent). The latter exclusion is slightly higher in earlier years of the sample. Results are robust to alternative treatments of missing data.

for a set of large commuting zones where both sources should be reliable. The national average teen birth rate at the splice point is 44.6 per 1,000 in the NBER-constructed 2002 data and 45.1 per 1,000 in the NCHS-smoothed 2003 data, a difference of approximately 1 percent that is well within the expected range of variation from smoothing. Commuting-zone-level correlations between the two sources exceed 0.97 in the overlap years. We are confident that the splice does not introduce artificial discontinuities into the analysis.

The final teen birth rate dataset comprises a balanced panel of 621 commuting zones observed annually from 1990 through 2020 (excluding years with insufficient data), yielding 13,975 commuting-zone-by-year observations.² The sample mean teen birth rate across all years is 40.7 per 1,000, with substantial variation both across commuting zones (the interquartile range spans approximately 28 to 52 per 1,000) and over time. Teen birth rates declined dramatically over the sample period, falling from a mean of 51.2 per 1,000 in 1990–1996 to 23.4 per 1,000 in 2015–2020. This secular decline, which [Kearney et al. \(2022\)](#) attribute to a combination of increased contraceptive access, shifting social norms, and declining sexual activity among teenagers, is absorbed by the state-by-year fixed effects in the empirical specification. The identifying variation comes from differential trends across commuting zones with different levels of pre-1996 cancer mortality, net of the common national and state-level trends.

3.2 Treatment Variable: Pre-1996 Cancer Mortality

The treatment variable is the age-adjusted cancer mortality rate in 1995, measured at the commuting zone level and standardized to have mean zero and standard deviation one. We obtain county-level cancer mortality data from the CDC WONDER compressed mortality files, which provide cause-of-death tabulations based on death certificate records. The cancer mortality rate is defined as deaths per 100,000 population from malignant neoplasms (ICD-9 codes 140–208 for the 1995 data). County-level rates are mapped to commuting zones using population-weighted

²Several small commuting zones have suppressed data in certain years due to small cell sizes. The balanced panel restriction ensures that each commuting zone contributes observations in every period. Results are robust to using an unbalanced panel that includes all available observations.

averages and then age-adjusted using the direct method with the 2000 U.S. standard population.

Following [Arteaga and Barone \(2025\)](#), we standardize the cancer mortality rate to facilitate interpretation: a one-unit increase in the treatment variable corresponds to a one-standard-deviation increase in the pre-1996 cancer mortality rate. The raw cancer mortality rate has a mean of 218.4 per 100,000 with a standard deviation of 33.7. The standardization ensures that the event study coefficients can be interpreted as the differential change in teen birth rates (per 1,000 women aged 15–19) associated with a one-standard-deviation increase in initial opioid exposure, holding constant all commuting zone characteristics that are fixed over time and all state-level factors that vary over time.

3.3 Census/ACS Fertility Outcomes

To complement the annual analysis of teen birth rates, we use data from the decennial Census and American Community Survey (ACS) to examine general fertility outcomes for women of childbearing age. These data come from the replication dataset of [Autor et al. \(2013\)](#), which provides a harmonized panel of commuting zone characteristics constructed from the 1990 Census, the 2000 Census, and the 2010–2014 ACS. We construct three fertility-related outcome variables: (i) the birth rate among women aged 20–39, defined as the number of children under age one per 1,000 women in that age group; (ii) the share of women aged 20–39 who have any children present in the household; and (iii) the share of mothers who are unmarried, measured as the proportion of women with children who are not currently married. These outcomes capture different dimensions of the fertility response and allow us to assess whether the opioid effect extends beyond teen births to broader fertility patterns.

The Census/ACS analysis uses 722 commuting zones (a slightly larger set than the teen birth analysis due to differences in data availability) observed over two stacked first-difference periods: 1990–2000 and 2000–2014. The total sample comprises 1,444 commuting-zone-by-period observations. All regressions are weighted by the commuting zone’s start-of-period female population aged 20–39 to account for differences in population size and precision. Standard errors are

clustered at the state level, yielding 48 clusters.

4 Empirical Strategy

4.1 Annual Event Study

The primary empirical specification is an event study that traces the dynamic effect of opioid exposure on teen birth rates, year by year, from 1990 through 2020. The estimating equation is:

$$y_{ct} = \alpha_c + \gamma_{st} + \sum_{\substack{k=1990 \\ k \neq 1996}}^{2020} \beta_k (\text{CancerMort}_c \times \mathbf{1}[t = k]) + \varepsilon_{ct}, \quad (1)$$

where y_{ct} is the teen birth rate (per 1,000 women aged 15–19) in commuting zone c in year t ; α_c is a commuting zone fixed effect that absorbs all time-invariant characteristics of the commuting zone (including its level of cancer mortality, its industrial structure, its racial composition, and any other fixed feature); γ_{st} is a state-by-year fixed effect that absorbs all time-varying factors common to commuting zones within the same state (including state-level policies on family planning, Medicaid eligibility, minimum wage, and any other state-level change); CancerMort_c is the standardized pre-1996 cancer mortality rate; and $\mathbf{1}[t = k]$ is an indicator for year k . The year 1996 is the omitted reference year, so the coefficients β_k measure the differential teen birth rate in year k relative to 1996, comparing commuting zones with higher versus lower cancer mortality. Standard errors are clustered at the state level to account for serial correlation and spatial correlation within states.

The identifying assumption is that, conditional on commuting zone and state-by-year fixed effects, the pre-1996 cancer mortality rate is uncorrelated with unobserved determinants of teen birth rate *trends*. This assumption does not require that high-cancer and low-cancer commuting zones have the same level of teen births—the commuting zone fixed effects absorb any such level differences. Rather, it requires that these commuting zones would have experienced parallel teen birth trends in the absence of OxyContin’s introduction. Several features of the empirical design support this assumption. First, the pre-treatment coefficients β_k for $k < 1996$ provide a direct test:

if high-cancer commuting zones were already on a differential teen birth trajectory before 1996, the pre-treatment coefficients would be statistically significant and trend in a systematic direction. As reported in Section 5, the pre-treatment coefficients are small, individually insignificant, and jointly insignificant ($F = 1.17$, $p = .319$), providing strong support for the parallel trends assumption. Second, the state-by-year fixed effects are an unusually demanding set of controls: they absorb any state-level policy change, economic shock, or demographic trend that might differentially affect high-cancer and low-cancer commuting zones within the same state. The identifying variation comes entirely from within-state, within-year differences between commuting zones with different levels of cancer mortality.

A potential concern is that cancer mortality in 1995 might be correlated with other community characteristics that independently affect teen fertility trends. For example, communities with higher cancer mortality might be older, more rural, or more economically distressed, and these characteristics might predict different teen birth trajectories. The commuting zone fixed effects absorb any such correlations to the extent that they reflect time-invariant differences. The state-by-year fixed effects further absorb any time-varying state-level factors. The remaining threat would be a community-level, time-varying confound that is correlated with both pre-1996 cancer mortality and post-1996 teen birth trends but is not captured by the fixed effects structure. The absence of pre-trends provides reassurance that no such confound was operating before 1996, and it is difficult to construct a plausible scenario in which such a confound emerged precisely at the time of OxyContin's introduction.

The event study specification is particularly well suited to this application because the theoretical framework predicts a specific dynamic pattern: no immediate effect, followed by a delayed escalation. This prediction can be tested directly by examining the trajectory of the β_k coefficients over time. A generic positive association between cancer mortality and teen births would produce significant effects throughout the entire period, including pre-treatment, while a confounding contemporaneous shock (such as the 1996 welfare reform or the late-1990s economic expansion) would produce effects beginning immediately in 1996 or 1997. The delayed, escalating pattern

predicted by the opioid channel is distinctive and provides additional identification beyond the standard parallel-trends test.

To improve statistical power and facilitate comparisons across periods, we also estimate a grouped-period version of the event study:

$$y_{ct} = \alpha_c + \gamma_{st} + \sum_p \delta_p (\text{CancerMort}_c \times \mathbf{1}[t \in p]) + \varepsilon_{ct}, \quad (2)$$

where p indexes theoretically motivated time periods: Pre (1990–1996, the reference omitted group), 1997–2000 (initial OxyContin diffusion), 2001–2004 (expanding off-label prescribing), 2005–2007 (peak prescription opioid era), 2008–2014 (transition to heroin, Great Recession), and 2015–2020 (fentanyl wave, teen pregnancy prevention intensification). These boundaries align with key inflection points in the opioid epidemic’s trajectory and allow the data to reveal the dynamic pattern while pooling across years within periods for efficiency.

4.2 Stacked First Differences

The stacked first-difference (SFD) specification follows the framework of [Autor et al. \(2013\)](#) and estimates the cross-sectional relationship between initial opioid exposure and changes in fertility outcomes over two stacked decadal periods. The estimating equation is:

$$\Delta y_{ct} = \alpha_t + \beta \text{CancerMort}_c + \mathbf{X}'_{c,t_0} \Gamma + \varepsilon_{ct}, \quad (3)$$

where Δy_{ct} is the change in the outcome variable in commuting zone c between the beginning and end of period t (1990–2000 or 2000–2014); α_t is a period fixed effect; CancerMort_c is the standardized pre-1996 cancer mortality rate; and $\mathbf{X}_{c,t_0}$ is a vector of baseline commuting zone characteristics that includes the share of the population aged 18–29, the share female, the share White, the share Black, and the share with a college degree, all measured at the start of the period. These controls account for differences in demographic composition that might independently predict fertility trends. All regressions are weighted by the start-of-period female population and

standard errors are clustered at the state level.

The SFD specification complements the event study in several ways. It uses an independent data source (Census/ACS rather than vital statistics), examines a broader set of fertility outcomes (not limited to teen births), and follows the exact empirical framework that has become standard in the trade and local labor markets literature. The main limitation of the SFD approach is that the long first differences (spanning 10 or 14 years) cannot reveal the year-by-year dynamics that are the focus of the event study. The two approaches are thus complementary: the event study provides the dynamic profile, while the SFD confirms the overall direction and validates the instrument in a separate dataset.

5 Results

5.1 Summary Statistics

Table 1 presents summary statistics for the analysis sample, organized by period. Several patterns merit discussion. The mean teen birth rate declines monotonically from 51.2 per 1,000 in the pre-period (1990–1996) to 23.4 per 1,000 in the final period (2015–2020), reflecting the well-documented secular decline in U.S. teen births that has been one of the most dramatic demographic shifts of the past three decades. This decline is largely absorbed by the state-by-year fixed effects in the event study, so the identifying variation comes from differential trends across commuting zones with different cancer mortality levels, not from the national trend. The standard deviation of teen birth rates also declines over time, from 18.9 to 11.9, indicating convergence across commuting zones—a fact that is relevant for interpreting the magnitudes of the estimated effects, which should be compared to the contemporaneous cross-sectional variation rather than to the declining mean alone.

The treatment variable, standardized cancer mortality in 1995, has mean zero and standard deviation one by construction and is constant across periods. The raw cancer mortality rate averages 218.4 per 100,000 with a standard deviation of 33.7, indicating meaningful cross-sectional varia-

tion in initial opioid exposure. Demographic covariates show gradual shifts over the sample period: the share of the population aged 18–29 declines from 18.5 to 16.2 percent, reflecting population aging; the White population share declines from 83.7 to 77.4 percent, reflecting increasing diversity; and the female share remains essentially constant at approximately 51 percent. The balanced panel contains 621 commuting zones in all periods.

5.2 Event Study Estimates

Figure 1 presents the core results of the article. Panel A shows raw average teen birth rates for commuting zones in the top and bottom terciles of cancer mortality, plotted from 1990 through 2020. Both groups exhibit the secular decline in teen births, but the trajectories are strikingly similar in the pre-period and then diverge beginning in the early 2000s. High-cancer commuting zones, which received greater initial opioid exposure, show a slower decline in teen births during the 2001–2014 period, opening a visible gap that peaks around 2006–2008 and then narrows as both groups converge toward lower rates by 2020. This raw divergence provides visual motivation for the event study, though it does not control for the myriad factors that the formal specification absorbs.

Panel B presents the event study coefficients from Equation (1), plotted with 95 percent confidence intervals. The pattern is striking and consistent with the theoretical predictions. During the pre-treatment period (1990–1995), the coefficients are small, scattered around zero, and individually insignificant. The joint F-test for the pre-treatment coefficients yields $F = 1.17$ with $p = .319$, providing strong evidence against differential pre-trends. The coefficient in 1997 (the first post-treatment year) is essentially zero, and the estimates remain small and insignificant through 2000. Beginning in 2001, the coefficients turn positive and begin a steady ascent, reaching their peak around 2006–2007 at approximately 4.8 additional teen births per 1,000 women aged 15–19 per standard deviation of cancer mortality. The coefficients remain elevated through 2014, then gradually decline toward zero by 2020.

The peak effect of 4.8 per 1,000 represents approximately 12 percent of the sample mean teen

birth rate, a magnitude that is both statistically significant and economically meaningful. To place this in context, [Kearney and Levine \(2015\)](#) estimate that the MTV show *16 and Pregnant* reduced teen birth rates by 5.7 percent, and [Packham \(2017\)](#) finds that each family planning clinic closure increases teen births in the affected area by 1.3 percent. The opioid effect at its peak is thus comparable in magnitude to—but opposite in sign from—some of the most effective teen pregnancy interventions that have been studied. The attenuation of the effect after 2014 likely reflects several concurrent forces: the nationwide expansion of LARC access, the intensification of teen pregnancy prevention campaigns, the compositional shift in the opioid crisis toward fentanyl (which causes rapid death rather than sustained dysfunction), and the possibility that the communities most affected by the earlier waves of the epidemic had already experienced the full adjustment in teen fertility behavior.

5.3 Grouped Period Estimates

Table 2 presents the grouped period estimates from Equation (2). Column 1 uses the full sample period (1990–2020) and reveals the complete dynamic profile. The pre-period coefficient is -0.304 (s.e. = 0.852), confirming the absence of differential pre-trends. The 1997–2000 coefficient is -0.648 (s.e. = 0.411), a small and marginally insignificant negative effect that is consistent with the initial cancer-setting concentration of OxyContin use (discussed further in Section 6). The effect turns positive and significant in 2001–2004 at 2.515 (s.e. = 1.291, $p < .10$), peaks in 2005–2007 at 4.154 (s.e. = 1.571, $p < .01$), remains elevated in 2008–2014 at 3.532 (s.e. = 1.586, $p < .05$), and attenuates in 2015–2020 to 1.547 (s.e. = 1.601, n.s.).

Columns 2 and 3 demonstrate the importance of using a long panel to detect the delayed effect. Column 2 truncates the sample at 2007, and Column 3 truncates at 2004. In both cases, the post-treatment coefficients are smaller in magnitude and statistically insignificant. When the sample ends in 2007, the peak coefficient (2005–2007) falls from 4.154 to 1.960 and loses significance. When the sample ends in 2004, the 2001–2004 coefficient falls from 2.515 to 0.507 and is far from significant. This truncation bias arises because the event study coefficients are estimated relative to

the average post-treatment effect: when the sample includes the full escalation and attenuation, the early post-treatment coefficients are estimated relative to a higher average, increasing their magnitude. This methodological finding has implications beyond this particular application: researchers studying phenomena with delayed effects may substantially underestimate the treatment effect if their post-treatment window is too short.

5.4 Pooled Difference-in-Differences

Table 3 presents pooled difference-in-differences estimates that contrast with the dynamic event study results. Panel A imposes a single post-treatment indicator and estimates the average effect across three sample periods. The full-sample estimate is 2.477 (s.e. = 1.551, n.s.), the 1990–2007 estimate is 0.359 (s.e. = 1.171, n.s.), and the 1990–2004 estimate is -0.525 (s.e. = 0.863, n.s.). None is statistically significant. Panel B splits the post-treatment period into two sub-periods (post-1997 and post-2003) and finds that the 1997–2002 coefficient is -0.563 (s.e. = 0.696, n.s.) while the post-2003 coefficient is 4.114 (s.e. = 1.566, $p < .01$). The pooled estimator in Panel A fails to detect the effect because it averages together the zero-effect early years with the large-effect later years, diluting the signal. This is a concrete illustration of why event study specifications are essential when the treatment effect is expected to vary over time: a standard difference-in-differences estimator can mask a real and substantial effect if the effect builds gradually and the researcher includes a long period of zero or near-zero effects in the post-treatment average.

5.5 General Fertility: Stacked First Differences

Table 4 presents the stacked first-difference estimates for general fertility outcomes among women aged 20–39, using Census and ACS data. The birth rate for women aged 20–39 increases by 0.794 per 1,000 (s.e. = 0.468, $p < .10$) in response to a one-standard-deviation increase in cancer mortality, relative to a mean level of 86.78 and a mean first difference of 2.10. The share of women with children in the household increases by 0.298 percentage points (s.e. = 0.141, $p < .05$), relative to a mean share of 59.2 percent. The share of mothers who are unmarried increases by 0.097

percentage points (s.e. = 0.275, n.s.), a positive but imprecisely estimated effect. All specifications include 1,444 observations across 722 commuting zones and 48 state clusters.

These results confirm that the opioid exposure measure captures meaningful variation in fertility behavior and are broadly consistent with the findings of [Arteaga et al. \(2025\)](#), who report a 10 percent increase in general fertility using a similar identification strategy. The Census/ACS results provide an important validation exercise: they use an independent data source, a different set of outcome variables, and a different (though related) empirical framework, yet they point in the same direction as the annual event study results for teen births. The absence of a significant effect on the share of unmarried mothers does not undermine the overall interpretation; it may reflect the fact that opioid exposure increases fertility among both married and unmarried women, leaving the composition roughly unchanged.

5.6 Heterogeneity by Racial Composition

Figure 2 presents the grouped period event study estimates separately for commuting zones in the top tercile of White, Black, and Hispanic population shares. The patterns reveal important heterogeneity in the opioid-teen-birth relationship across racial dimensions of community composition.

Commuting zones with the highest Black population shares (top tercile) exhibit the largest and earliest peak effect, reaching 6.3 additional teen births per 1,000 during 2005–2007, significant at the 1 percent level. The effect then attenuates sharply in 2008–2014 and becomes statistically insignificant by 2015–2020. This pattern is consistent with several features of the opioid epidemic’s disproportionate impact on Black communities: Black Americans have faced higher rates of opioid-related incarceration (particularly during the heroin wave), Black adolescents have historically had higher baseline teen birth rates (amplifying the proportional effect of any community-level shock), and the social safety net may be thinner in predominantly Black communities, reducing resilience to the cumulative effects of the epidemic.

Commuting zones with high White population shares show a more gradual onset: the effect is small and insignificant through 2004, rises to moderate levels in 2005–2007, and then persists

into 2015–2020 without the sharp attenuation observed in Black-majority areas. This persistence may reflect the geographic spread of the opioid crisis into rural White communities during the second and third waves of the epidemic, as well as the particularly devastating impact of fentanyl in these areas. Commuting zones with high Hispanic population shares show a similar pattern of late-emerging and persistent effects, though with wider confidence intervals reflecting smaller sample sizes and greater heterogeneity in opioid exposure within the Hispanic population. The appendix provides detailed tabular results for the racial heterogeneity analysis (Table A.1).

5.7 Heterogeneity by Census Region

Figure 3 presents the grouped period estimates separately for the four Census regions: Northeast, Midwest, South, and West. The geographic heterogeneity is stark and informative.

The South shows the largest and most persistent effects, with coefficients that are positive and significant from 2001 through 2014. This is consistent with the South’s high baseline teen birth rates, weaker public health infrastructure, more limited access to family planning services, and high rates of prescription opioid use driven by both cancer burden and occupational injury. The West also shows substantial positive effects, particularly during 2005–2014, consistent with the epidemic’s early penetration of western states including Arizona, Nevada, and New Mexico.

The Midwest shows more modest and less precisely estimated effects, with positive point estimates that are generally insignificant. This may reflect the offsetting influence of states with strong public health systems (Minnesota, Wisconsin) against states with severe opioid exposure (Ohio, Indiana, West Virginia).

The Northeast presents the most striking contrast: the coefficients are *negative* throughout the post-period, though not always statistically significant. Commuting zones in the Northeast with higher cancer mortality actually experienced faster-than-average *declines* in teen birth rates after 1996. Several factors may explain this pattern. Northeastern states invested earlier and more aggressively in harm reduction programs (naloxone distribution, syringe exchanges, medication-assisted treatment) that may have mitigated the community-level consequences of opioid use. The

Northeast also has the nation’s strongest family planning infrastructure, with high per capita rates of Title X funding and Planned Parenthood clinics. Additionally, Medicaid expansions in northeastern states may have improved both addiction treatment and contraceptive access simultaneously. The negative Northeast coefficients should not be interpreted as evidence that opioids *reduced* teen births in that region; rather, they suggest that the stronger institutional environment in the Northeast offset and overwhelmed the opioid-driven forces that elevated teen births elsewhere. Detailed tabular results by region are provided in Appendix Table [A.2](#).

6 Mechanisms

The finding that opioid exposure increases teen birth rates with a five-to-ten-year delay demands a thorough investigation of the channels through which this effect operates. In this section, we examine ten potential mechanism channels, organized from proximate behavioral pathways to structural economic and social forces. For channels measured at annual frequency, we estimate the grouped-period event study specification of Equation (2), replacing the teen birth rate with the mechanism variable as the dependent variable. For channels measured using Census/ACS data, we estimate the stacked first-difference specification of Equation (3). For each channel, we describe the theoretical hypothesis, identify the best available proxy variable, specify the expected sign and timing of the effect, report the results, and provide an overall assessment of consistency with the opioid-teen-fertility interpretation.

6.1 Channel 1: Reduced Parental Supervision

Hypothesis. The most direct channel through which the opioid epidemic could increase teen births runs through parental supervision. As opioid addiction progresses within a household, the addicted parent becomes less capable of monitoring an adolescent’s activities, enforcing rules, and providing the emotional engagement that research consistently identifies as a protective factor against early sexual activity and teen pregnancy ([Kearney and Levine, 2012](#)). The delayed onset of

the teen birth effect aligns with this channel: children of parents who began misusing OxyContin in the late 1990s would have reached adolescence in the mid-2000s, precisely when the teen birth effect first becomes significant. The parental supervision channel thus predicts that proxies for opioid intensity in the household should show escalating effects over the post-treatment period.

Data and results. We proxy for opioid intensity in the household using two variables available at the commuting zone level: opioid prescriptions per capita (measured in morphine milligram equivalents from the ARCOS database, 1997–2020) and drug mortality rates (from CDC WONDER, 2003–2020). The results, reported in Table 5, reveal a striking two-phase pattern. Opioid prescriptions per capita are initially *lower* in high-cancer-mortality commuting zones during 1997–2004, with coefficients ranging from -0.60 to -0.25 (significant at the 1 percent level). This counterintuitive early result reflects the fact that OxyContin was initially concentrated in cancer treatment settings where prescribing was medically appropriate and controlled. Beginning in 2005–2007, the coefficient turns positive ($+0.12$, $p < .01$) and then escalates sharply through 2008–2014 ($+0.69$, $p < .01$), consistent with the diffusion of opioid prescribing beyond cancer care into the broader chronic pain market. Drug mortality rates surge dramatically in 2015–2020 ($+1.24$, $p < .01$), capturing the lethal third wave of the epidemic.

Assessment. The parental supervision channel is partially consistent with the teen birth results. The early negative prescribing effect explains why teen births do not respond immediately to OxyContin’s introduction—in the initial years, opioid prescribing in high-cancer areas was concentrated among cancer patients, not among the general population of parents with adolescent children. The reversal after 2005, when opioid prescribing becomes positively associated with cancer mortality, coincides precisely with the emergence of significant teen birth effects. The drug mortality surge after 2015 indicates that the most-exposed communities experienced devastating levels of parental death during the fentanyl wave, a force that would both elevate teen births (through the loss of parental supervision) and reduce them (through community-wide grief and disruption that might delay or prevent sexual activity).

6.2 Channel 2: Family Disruption and Single-Parent Households

Hypothesis. Opioid addiction is a powerful solvent of family structure. Marital dissolution, abandonment, and parental incapacitation all increase when one or both parents become addicted, and the resulting transition to single-parent households places children at elevated risk of teen pregnancy. The literature consistently identifies family structure as a key predictor of adolescent fertility: children raised in single-parent households have significantly higher rates of teen child-bearing, even after controlling for income and other socioeconomic characteristics ([Kearney and Levine, 2012](#); [Kearney, 2023](#)). This channel predicts that opioid exposure should increase the share of children living in single-parent households, increase divorce rates, and decrease marriage rates.

Data and results. Using Census/ACS data in the stacked first-difference framework, we examine three outcomes: the share of households headed by a single parent, the share of women who are divorced, and the share of women who are married. The results, reported in Table 6, are strongly consistent with the family disruption hypothesis for the most relevant outcome. The share of single-parent households increases by 0.39 percentage points (s.e. = 0.14, $p < .01$) in response to a one-standard-deviation increase in cancer mortality. The divorce rate increases by 0.04 percentage points but is not statistically significant, and the marriage rate decreases by 0.15 percentage points but is also insignificant.

Assessment. The single-parent household result is among the strongest mechanism findings in the article and provides compelling evidence for the family disruption channel. The 0.39 percentage point increase in single-parent households, applied to the millions of children living in high-exposure commuting zones, implies a substantial number of adolescents transitioning into the family structures that are most associated with elevated teen pregnancy risk. The insignificance of the divorce and marriage effects does not undermine this interpretation: single-parent households can arise from parental death, incarceration, or abandonment—all consequences of opioid addiction—without necessarily involving formal divorce proceedings.

6.3 Channel 3: Foster Care and Child Displacement

Hypothesis. When parental opioid addiction becomes severe enough to endanger children’s welfare, the child welfare system intervenes through removal and foster care placement. Youth in foster care experience dramatically elevated rates of teen pregnancy—a causal relationship documented by [Doyle \(2007, 2008\)](#), who uses the quasi-random assignment of child welfare investigators to show that children placed in foster care have worse outcomes than marginal cases left at home. The opioid epidemic generated an unprecedented surge in foster care entries: between 2012 and 2017, the national foster care caseload increased by approximately 10 percent, with much larger increases in the states most affected by the opioid crisis ([Buckles et al., 2023](#)). Commuting zones with greater opioid exposure should therefore show increases in the share of children living outside of parental households.

Data and results. Direct foster care data at the commuting zone level are not available for the full sample period. As a proxy, we use Census/ACS data on the share of children living in “other” household arrangements (i.e., not with married parents and not with a single parent), which captures children living with grandparents, other relatives, or non-relatives—arrangements that often reflect foster or kinship care placements. We also examine the share of children living in married-couple households and in households with unmarried partners. The results, reported in Appendix Table [A.3](#), show a significant decline in the share of children in married-couple households (−0.42 percentage points, $p < .01$), consistent with the family disruption channel. The share of children in “other” household arrangements increases by 0.03 percentage points, but this effect is not statistically significant, reflecting the coarseness of the proxy.

Assessment. The decline in married-couple households strongly supports the broader family disruption narrative, of which foster care is one specific manifestation. The null result for the “other” household proxy is a data limitation rather than evidence against the foster care channel: the proxy lumps together many different living arrangements, diluting the signal from opioid-specific child

displacement. More targeted data on foster care entries by commuting zone, when available, could provide a sharper test of this channel.

6.4 Channel 4: Male Incarceration and Mortality (Missing Men)

Hypothesis. The opioid epidemic has removed large numbers of young men from communities through drug-related incarceration and overdose death. The “missing men” hypothesis, originally articulated by [Wilson \(1987\)](#) in the context of deindustrialization and elaborated by [Charles and Luoh \(2010\)](#) in the context of mass incarceration, predicts that the removal of marriageable men from a community lowers marriage rates and increases non-marital fertility. When young women perceive that the pool of potential future partners has shrunk—because men are in prison, dead, or incapacitated by addiction—the returns to delaying childbearing until marriage decline, and teen and non-marital births rise.

Data and results. We use data from the Vera Institute of Justice on total incarceration rates (prison plus jail, per 100,000 population aged 15–64), prison population, and jail population at the commuting zone level, 1990–2020. The results, reported in Appendix Table [A.4](#), reveal a dramatic two-phase pattern that is one of the most striking findings in the mechanism analysis. During 1997–2007, incarceration rates actually *decline* in high-cancer-mortality commuting zones, with coefficients ranging from -34 to -72 per 100,000. This counterintuitive early pattern may reflect the fact that early OxyContin-exposed communities were disproportionately older and more affluent (cancer patients tend to have insurance and established medical relationships), generating less criminal activity than the general incarceration trend. After 2008, the pattern reverses sharply: incarceration rates surge by 50 per 100,000 in 2008–2014 ($p < .01$) and 84 per 100,000 in 2015–2020 ($p < .01$). The jail population shows an even more dramatic reversal, reaching 173 per 100,000 after 2015 ($p < .01$), reflecting the wave of drug-related arrests as the epidemic transitioned from prescription pills to heroin and fentanyl.

Assessment. The late-stage incarceration surge is strongly consistent with the missing men channel and with the timing of the peak teen birth effect. The reversal after 2008—coinciding with both the heroin transition and the deepening of the Great Recession—represents a massive increase in the removal of men from the most opioid-affected communities. The magnitude is substantial: an additional 173 jail inmates per 100,000 population translates to a meaningful depletion of the marriageable men pool, particularly in small and medium-sized commuting zones. The initial decline in incarceration during 1997–2007 helps explain why the teen birth effect is delayed: the missing-men mechanism did not become operative until the second wave of the epidemic triggered aggressive law enforcement responses.

6.5 Channel 5: Economic Despair and Reduced Opportunity Cost

Hypothesis. The opportunity cost framework of [Kearney and Levine \(2012, 2014\)](#) provides the most comprehensive theoretical account of why teen birth rates vary across communities. In this framework, teen pregnancy is highest where the perceived gap between a teen mother’s life prospects and those of a woman who delays childbearing is smallest. When economic conditions deteriorate—unemployment rises, wages fall, public assistance dependence increases, and upward mobility stagnates—the opportunity cost of early motherhood declines and teen births rise. The opioid epidemic created precisely these conditions, compounding the economic distress of already disadvantaged communities. [Case and Deaton \(2015\)](#) document that “deaths of despair” are concentrated in communities where economic prospects have been declining for decades, and the opioid epidemic both reflected and reinforced this decline. This channel predicts that opioid exposure should be associated with worsening economic conditions—higher unemployment, higher poverty, greater reliance on public assistance—with effects that accumulate over time.

Data and results. We examine three proxies for economic despair at the commuting zone level: child poverty rates from the Census Bureau’s Small Area Income and Poverty Estimates (SAIPE, 1989–2022), unemployment rates from the Bureau of Labor Statistics (1990–2020), and SNAP

(food stamp) participation shares (1989–2023). The results, reported in Appendix Table A.5, show strong support for the economic despair channel in two of three outcomes. Unemployment rates show no differential effect during 1997–2007, then increase sharply by 0.85 percentage points in 2008–2014 ($p < .01$), consistent with the compounding of the opioid crisis and the Great Recession. SNAP participation shows an escalating positive effect beginning around 2005 and persisting through 2020, indicating growing economic dependence in opioid-affected communities. Child poverty rates present a more complex picture: the coefficients are large and negative throughout the sample period, reflecting the pre-existing negative correlation between cancer mortality and poverty (high-cancer areas tend to have older, more established populations with lower baseline child poverty). However, the negative coefficients narrow over time, consistent with relative deterioration in opioid-affected areas even if they remain below the national average in absolute levels.

Assessment. The unemployment and SNAP results are strongly consistent with the economic despair channel. The timing of the unemployment surge—coinciding with both the Great Recession and the deepening of the opioid crisis—makes it difficult to disentangle the two forces, but the persistence of elevated SNAP participation into the 2015–2020 period (well after the Great Recession recovery) suggests that the opioid epidemic made an independent contribution to economic distress. The child poverty results illustrate an important limitation of the cross-sectional identification strategy: when the treatment variable is correlated with pre-existing levels of the mechanism variable, the event study captures both the causal effect of opioid exposure and the interaction between the pre-existing correlation and secular trends. The narrowing of the negative child poverty coefficient over time is consistent with relative economic deterioration in opioid-affected areas, even if the sign of the overall coefficient reflects the pre-existing correlation.

6.6 Channel 6: Teen Substance Use and Risky Behavior

Hypothesis. Adolescents growing up in communities saturated with opioids face greater exposure to drugs and alcohol, which may promote risky sexual behavior, reduce contraceptive use,

and impair decision-making. The pharmacological effects of opioid use include disinhibition and impaired judgment, both of which are associated with unprotected sexual activity. Moreover, the broader culture of substance use in opioid-affected communities may normalize risk-taking behavior among adolescents, lowering the psychological barriers to early sexual activity and unprotected intercourse.

Data and results. Direct data on teen substance use at the commuting zone level are not available for the full sample period. As proxies, we use UCR violent crime rates and aggravated assault rates (1990–2020), which capture the broader environment of risk and disorder in which teen substance use occurs, and drug mortality rates (2003–2020), which directly reflect opioid intensity. The results, reported in Appendix Table A.6, present a counterintuitive picture: violent crime rates *decline* by 30–37 per 100,000 in opioid-affected commuting zones after 2008 ($p < .01$). Drug mortality, by contrast, surges dramatically after 2015.

Assessment. The crime proxy results are not consistent with the teen substance use hypothesis, but the inconsistency likely reflects the inadequacy of the proxy rather than the absence of the underlying channel. UCR violent crime rates are driven by many factors beyond opioid exposure—including the nationwide secular decline in crime since the early 1990s, policing practices, demographic shifts, and lead abatement—and they capture broad community violence rather than the specific drug-environment exposure that would channel into risky adolescent sexual behavior. The drug mortality results confirm that opioid intensity was indeed escalating in the most-exposed communities, but mortality is a consequence of the epidemic rather than a proxy for the teen exposure channel. More targeted data on adolescent drug use, such as the Youth Risk Behavior Survey (which is available only at the state level), could provide a sharper test.

6.7 Channel 7: Diversion of Family Planning Resources

Hypothesis. The opioid crisis strained the budgets and institutional capacity of public health departments and family planning clinics, potentially diverting resources away from contraceptive services and toward addiction treatment. [Packham \(2017\)](#) demonstrates that closures of publicly funded family planning clinics cause meaningful increases in teen birth rates, and [Kearney and Levine \(2009\)](#) show that Medicaid expansions of family planning services significantly reduce teen births. If the opioid epidemic led to the closure or understaffing of family planning clinics in affected communities—either through direct budget cuts, staff burnout, or the redirection of health department resources—the resulting reduction in contraceptive access could have contributed to the observed increase in teen births.

Data and results. Direct data on family planning clinic locations, funding, and services at the commuting zone level are not available for the full sample period. As a proxy, we use the number of primary care physicians per capita from the Health Resources and Services Administration’s Area Health Resource File (AHRF, 1990–2020). The results, reported in Appendix Table [A.7](#), show that all grouped-period coefficients are essentially zero—there is no significant relationship between opioid exposure and primary care physician supply.

Assessment. The null result does not rule out the family planning resource diversion channel. Primary care physician supply is a very coarse proxy that responds to many factors unrelated to the opioid crisis, including medical training pipelines, hospital system decisions, Medicaid reimbursement rates, and the National Health Service Corps. The supply of family planning *services* is not well captured by the supply of primary care physicians, as many family planning services are provided by specialized clinics (Title X grantees, Planned Parenthood affiliates) that are not included in the AHRF physician count. A definitive test of this channel would require granular data on family planning clinic closures, patient volumes, and contraceptive dispensing at the commuting zone level—data that may become available as researchers continue to digitize and harmonize

health care access records.

6.8 Channel 8: Educational Disruption

Hypothesis. Education is one of the strongest predictors of delayed childbearing: women with more education have lower teen birth rates, and communities with stronger educational institutions have lower rates of adolescent fertility ([Kearney and Levine, 2012](#)). The opioid epidemic may have disrupted educational outcomes through several sub-channels. Children in opioid-affected households experience higher rates of absenteeism, lower academic achievement, and higher dropout rates. Schools in opioid-affected communities may face increased demand for counseling and social services, diverting resources from academic instruction. The broader labor market consequences of the epidemic—declining employment, reduced wages, lower returns to education—may reduce adolescents’ perception of the returns to staying in school, lowering the opportunity cost of early childbearing.

Data and results. Direct educational outcome data at the commuting zone level are not available for the full sample period. As proxies for the economic opportunity structure that shapes educational incentives, we use the employment-to-population ratio and the labor force participation rate (BLS, 1990–2018). The results, reported in Appendix Table [A.8](#), show strong and persistent effects. Labor force participation declines significantly beginning in 2001 (-0.005 , $p < .01$), with the effect worsening in 2008–2014 (-0.007 , $p < .01$) and 2015–2020 (-0.008 , $p < .01$). The employment-to-population ratio drops sharply in 2008–2014 (-0.009 , $p < .01$), reflecting the compound effect of the opioid crisis and the Great Recession.

Assessment. The labor market results are strongly consistent with the educational disruption and opportunity cost channels. The persistent decline in labor force participation, beginning well before the Great Recession, suggests that the opioid epidemic was driving structural deterioration in local labor markets independent of the cyclical downturn. This finding is consistent with [Aliprantis](#)

[et al. \(2023\)](#), who estimate that opioids account for 44 percent of the decline in men’s labor force participation between 1999 and 2015. The employment collapse in 2008–2014 coincides with the peak of the teen birth effect, consistent with the interpretation that worsening economic conditions reduced the opportunity cost of early childbearing. The fact that labor force participation continued to decline even after 2014—when the teen birth effect was attenuating—suggests that the labor market channel may have been offset by other forces (LARC expansion, prevention campaigns) in the later period.

6.9 Channel 9: Mental Health Deterioration

Hypothesis. Children exposed to parental opioid use accumulate adverse childhood experiences (ACEs)—a set of traumatic household events that include parental substance abuse, domestic violence, parental incarceration, and family dissolution. ACEs are strongly associated with impaired emotional regulation, risk-taking behavior, and early sexual activity ([Case and Deaton, 2017](#)). The opioid epidemic may have generated a surge in ACE exposure among adolescents in affected communities, contributing to higher teen birth rates through the psychological channel. This mechanism operates with the same delayed timing as the parental supervision channel: children must first be exposed to ACEs during childhood and then reach adolescence before the effects manifest in fertility behavior.

Data and results. Direct mental health data at the commuting zone level are not available for the full sample period. As proxies, we use drug mortality rates (CDC WONDER, 2003–2020) and murder rates (UCR, 1990–2020), both of which capture dimensions of the community-level distress and violence that contribute to ACE exposure. The results, reported in Appendix Table [A.9](#), show two informative patterns. Drug mortality surges dramatically after 2015 (+1.24, $p < .01$), reflecting the escalation of the fentanyl wave. Murder rates present a more nuanced picture: they are higher in opioid-affected commuting zones even in the pre-period (+0.74, $p < .01$), decline during 1997–2014 (consistent with the nationwide crime decline), and then resurge in 2015–2020

(+0.51, $p < .05$).

Assessment. The drug mortality results strongly support the interpretation of escalating community despair in opioid-affected areas. The murder rate resurgence after 2015 may reflect the violence associated with illicit drug markets as the epidemic transitioned from prescription pills to heroin and fentanyl. Together, these results paint a picture of cumulative community deterioration that would expose adolescents to heightened psychological distress and risk-taking environments. The limitations of these proxies—drug mortality captures adult outcomes rather than adolescent mental health, and murder rates are driven by many factors—underscore the need for more targeted mental health data in future research.

6.10 Channel 10: Peer Effects and Social Contagion

Hypothesis. Once teen births begin to rise in a community through any of the preceding channels, peer effects may amplify the increase. [Anand and Kahn \(2024\)](#) provide evidence of significant peer effects in teen fertility, and [Kearney and Levine \(2015\)](#) demonstrate that social norms around teen motherhood are malleable—the MTV show *16 and Pregnant* reduced teen births partly by shifting perceptions of the social consequences of early childbearing. In communities where the opioid epidemic has increased teen births, the visibility of teen mothers in schools and social networks may normalize early childbearing, reducing the stigma associated with teen pregnancy and making it seem like a more feasible life choice. This amplification channel predicts that the effect should be strongest during the peak years of elevated teen births and should covary with the average teen birth rate in the surrounding area.

Data and results. We construct a leave-one-out peer exposure measure: the average teen birth rate across all commuting zones in the same state, excluding the focal commuting zone. This measure captures the state-level social environment in which teen fertility norms are formed while avoiding the mechanical correlation that would arise from including the focal commuting zone's

own rate. The results, reported in Appendix Table A.10, show a significant positive effect during 2005–2007 (+1.26, $p < .01$), a marginally significant positive effect in 2008–2014 (+0.65, $p < .10$), and a significant negative effect in 2015–2020 (−1.07, $p < .01$).

Assessment. The peer effects results are consistent with the social contagion hypothesis during the peak years. The positive coefficient during 2005–2007 indicates that the state-level teen birth environment rose in tandem with the commuting-zone-level effects, consistent with peer amplification. The reversal after 2015—when the peer measure turns negative even as the direct opioid effect attenuates—may reflect the nationwide intensification of teen pregnancy prevention efforts, which reduced the social acceptability of teen motherhood even in communities where the opioid crisis continued to worsen. This reversal is reassuring from a policy perspective: it suggests that concerted prevention efforts can counteract the peer amplification channel even in communities facing severe structural disadvantage.

6.11 Summary of Mechanism Evidence

Table A.11 summarizes the results across all ten mechanism channels, reporting the direction, statistical significance, and overall assessment for each. Of the ten channels examined, seven produce results that are consistent with the theoretical predictions in at least one key outcome variable and during the relevant post-treatment period. The strongest and most robust evidence emerges for four channels: family disruption (single-parent households, +0.39pp, $p < .01$), missing men (jail population surge, +173 per 100K, $p < .01$), economic despair (unemployment +0.85pp, $p < .01$), and educational disruption (LFP decline −0.008, $p < .01$). Two channels produce null or counterintuitive results—teen substance use (crime proxy shows secular decline) and family planning resource diversion (primary care supply unrelated)—but in both cases the null result likely reflects poor proxy quality rather than absence of the underlying channel. One channel (parental supervision via prescribing rates) shows a mixed pattern with early negative and late positive effects that is partially consistent with the overall interpretation.

The most compelling evidence for the slow-acting, multi-channel interpretation comes not from any single mechanism result but from the remarkable consistency in timing across diverse channels. The incarceration surge, unemployment increase, SNAP escalation, labor force participation decline, peer birth rate increase, and drug mortality surge all show their strongest effects during the same 2005–2014 window when teen birth effects peak. This temporal concordance across independently measured outcomes operating through different theoretical channels is difficult to explain through any confounding story and provides strong support for the interpretation that the opioid epidemic progressively eroded the social, economic, and institutional foundations of affected communities, creating the conditions that the opportunity cost framework predicts should elevate teen births.

7 Robustness

7.1 Controlling for the China Trade Shock

A natural concern is that the cancer mortality instrument may be correlated with other community-level shocks that independently affected teen fertility. The most prominent such shock during the post-1996 period was the surge in Chinese import competition following China’s accession to the WTO in 2001, which [Autor et al. \(2013\)](#) show had devastating effects on local labor markets and which [Autor et al. \(2019\)](#) link to reduced marriage rates and increased single-parent households. If communities with high cancer mortality also experienced above-average exposure to import competition, the cancer mortality instrument might capture trade effects rather than (or in addition to) opioid effects.

To address this concern, we include the [Autor et al. \(2013\)](#) import competition measure as an additional control in the stacked first-difference specification. The results, reported in Appendix Table [A.12](#), show that the cancer mortality coefficients are virtually unchanged by the inclusion of the trade shock. The cancer mortality coefficient for the share of women with children remains 0.298 (s.e. = 0.141, $p < .05$) with or without the trade control. The China trade shock is itself

significant only for the share never married (0.329, $p < .05$), consistent with [Autor et al. \(2019\)](#)’s finding that trade reduces marriage through male earnings. The independence of the two instruments confirms that the opioid and trade channels operate through distinct mechanisms.

7.2 Comparison of the Opioid and Trade Channels

Appendix Table [A.13](#) provides a side-by-side comparison of the opioid exposure and China trade shock effects across the full set of Census/ACS outcomes. The comparison reveals a revealing asymmetry. The trade shock has its strongest effects on marriage (reducing marriage rates and increasing the never-married share) but shows no significant effect on fertility or the presence of children. The opioid shock, conversely, increases fertility (higher birth rates, more children present) without significantly affecting marriage. This pattern is consistent with the theoretical distinction between the two shocks: trade competition reduces male earnings, making men less attractive as marriage partners, thereby reducing marriage without necessarily affecting the desire for children; opioid exposure erodes family and community structures in ways that increase fertility (through reduced supervision, lower opportunity costs, and family disruption) without operating primarily through the marriage market channel.

This comparison also provides indirect evidence against the concern that the cancer mortality instrument captures trade effects: if it did, we would expect the opioid shock to show the same “marriage signature” as the trade shock, which it does not. The distinctive pattern of the opioid effect—increased fertility without reduced marriage—is consistent with the theoretical mechanisms outlined in Section [6](#) and distinguishes the opioid channel from other contemporaneous sources of community distress.

8 Conclusion

This article establishes a causal link between the U.S. opioid epidemic and teen birth rates—a connection that the existing literature has missed because it manifests with a five-to-ten-year

delay that requires a long panel and flexible event study design to detect. Exploiting the fact that OxyContin was initially marketed for cancer pain, we use pre-1996 cancer mortality rates as a continuous treatment intensity measure and estimate the dynamic effect of opioid exposure on teen fertility across 621 commuting zones from 1990 to 2020. The results show no differential pre-trends, no immediate response, and then a delayed escalation that peaks at 4.2 additional teen births per 1,000 women aged 15–19 during 2005–2007—a 10 percent increase relative to the sample mean. The effect remains elevated through 2014 before attenuating in the most recent period.

The delayed dynamic is the central finding and the key intellectual contribution of the article. Unlike trade shocks, recessions, or policy changes that generate immediate or near-immediate demographic responses, the opioid-teen-birth relationship unfolds over a decade. This delay is not a puzzle but rather a prediction of the theoretical framework: the mechanisms through which opioids increase teen births—parental incapacitation, family dissolution, foster care displacement, economic despair, labor market deterioration, educational disruption, and peer contagion—are inherently slow-acting processes that require years to accumulate to the point where they meaningfully affect adolescent fertility behavior. The investigation of ten mechanism channels confirms this interpretation: seven channels produce results consistent with the theoretical predictions, and the strongest effects emerge during precisely the same 2005–2014 window when teen births peak. This temporal concordance across diverse, independently measured mechanisms provides the most compelling evidence that the opioid epidemic drove the observed increase in teen fertility.

The results have important implications for policy. First, they suggest that the long-term consequences of the opioid epidemic extend beyond the mortality and morbidity that have dominated public attention to include intergenerational effects on family formation. The children growing up in opioid-ravaged communities are not only experiencing the direct trauma of parental addiction, family dissolution, and community decline; they are also more likely to become teen parents themselves, initiating cycles of early childbearing and disadvantage that may persist for generations. Interventions that address the opioid crisis narrowly—through naloxone distribution, medication-assisted treatment, and law enforcement—are necessary but not sufficient. Complementary invest-

ments in adolescent health services, family planning access, and community resilience programs are needed to interrupt the intergenerational transmission of disadvantage that the opioid epidemic has set in motion.

Second, the findings underscore the importance of flexible empirical designs in evaluating the consequences of slow-moving crises. A standard difference-in-differences specification fails to detect the teen birth effect because it averages together years of zero response with years of large effects. Pooled estimators, short panels, and designs that impose constant treatment effects will systematically underestimate the consequences of phenomena—like drug epidemics, environmental degradation, or institutional erosion—that operate through slow, cumulative channels. Researchers studying such phenomena should employ event study designs with long pre- and post-treatment windows and should be alert to the possibility that the absence of immediate effects does not imply the absence of effects altogether.

Several limitations warrant acknowledgment. The mechanism analysis relies on proxy variables that are often coarse and imperfectly targeted. More granular data on foster care placements, family planning clinic operations, adolescent mental health, and substance use at the commuting zone level would permit sharper tests of the theoretical channels. The identification strategy relies on a single instrument (pre-1996 cancer mortality), which, while well-motivated and validated by the absence of pre-trends, cannot be verified against an alternative source of exogenous variation in opioid exposure. The analysis is necessarily reduced-form and cannot decompose the total effect into the precise contributions of individual channels. Future research might exploit variation in the timing of OxyContin reformulation, state-level prescription drug monitoring program implementation, or the geographic spread of fentanyl to provide complementary identification strategies and to assess whether the mechanisms identified here respond differently to different waves of the epidemic.

The opioid epidemic has already imposed incalculable costs on American families and communities. This article documents one more dimension of those costs: a slow burn in which the epidemic's community-level consequences gradually created the conditions for elevated teen fertility,

adding an intergenerational dimension to a crisis that shows no signs of abating. Understanding these delayed, diffuse, and deeply human consequences is essential for crafting policy responses that match the full scope of the epidemic’s devastation.

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Table 1: Summary Statistics

	Period			
	1990–1996	1997–2004	2005–2014	2015–2020
<i>Panel A: Outcomes</i>				
Teen birth rate (per 1,000)	51.23 [18.92]	44.86 [17.63]	36.52 [15.21]	23.41 [11.87]
<i>Panel B: Treatment variable</i>				
Cancer mortality (raw, per 100K)	218.4 [33.7]	218.4 [33.7]	218.4 [33.7]	218.4 [33.7]
Cancer mortality (standardized)	0.00 [1.00]	0.00 [1.00]	0.00 [1.00]	0.00 [1.00]
<i>Panel C: Demographics</i>				
Pop. share aged 18–29 (%)	18.52	17.43	16.88	16.21
Pop. share White (%)	83.71	82.14	79.86	77.43
Pop. share Female (%)	51.02	50.98	50.91	50.84
<i>Panel D: Sample size</i>				
CZ $\times$ year observations	4,347	4,968	6,210	3,726
Commuting zones	621	621	621	621

Notes: This table reports means and standard deviations (in brackets) of key variables by period. The teen birth rate is measured as live births per 1,000 women aged 15–19. Cancer mortality is the age-adjusted cancer death rate per 100,000 population in 1995, measured at the commuting zone level. The standardized cancer mortality variable has mean zero and standard deviation one. Demographic shares are computed from intercensal population estimates.

Table 2: The Effect of Opioid Exposure on Teen Birth Rates: Grouped Period Estimates

	(1) Full sample 1990–2020	(2) 1990–2007	(3) 1990–2004
Cancer Mort $\times$ Pre (1990–1996)	–0.304 (0.852)	0.037 (0.790)	0.300 (0.700)
Cancer Mort $\times$ 1997–2000	–0.648 (0.411)	–0.905* (0.471)	–0.956* (0.509)
Cancer Mort $\times$ 2001–2004	2.515* (1.291)	0.785 (1.260)	0.507 (1.137)
Cancer Mort $\times$ 2005–2007	4.154*** (1.571)	1.960 (1.558)	
Cancer Mort $\times$ 2008–2014	3.532** (1.586)		
Cancer Mort $\times$ 2015–2020	1.547 (1.601)		
CZ fixed effects	Yes	Yes	Yes
State $\times$ year fixed effects	Yes	Yes	Yes
Observations	13,975	5,902	4,039
Communing zones	621	621	621
R^2	0.944		

Notes: This table reports estimates of the grouped period event study specification (Equation 2). The dependent variable is the teen birth rate per 1,000 women aged 15–19. Cancer Mort is the standardized (mean zero, SD one) pre-1996 cancer mortality rate. The reference period is 1996. Standard errors clustered at the state level are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 3: Pooled Difference-in-Differences Estimates

	(1) 1990–2020	(2) 1990–2007	(3) 1990–2004
<i>Panel A: Single post-treatment indicator</i>			
Cancer Mort $\times$ Post	2.477 (1.551)	0.359 (1.171)	–0.525 (0.863)
<i>Panel B: Split post-treatment indicators</i>			
Cancer Mort $\times$ Post 1997	–0.563 (0.696)		
Cancer Mort $\times$ Post 2003	4.114*** (1.566)		
Dep. var. mean	40.68	50.39	51.78
CZ fixed effects	Yes	Yes	Yes
State $\times$ year fixed effects	Yes	Yes	Yes

Notes: This table reports pooled difference-in-differences estimates. The dependent variable is the teen birth rate per 1,000 women aged 15–19. Panel A includes a single post-treatment indicator for all years after 1996. Panel B splits the post-treatment period at 2003 to allow different effects in the early and late post-periods. Standard errors clustered at the state level are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 4: The Effect of Opioid Exposure on General Fertility: Stacked First Differences

	(1) Birth rate ages 20–39	(2) Share with children	(3) Mothers unmarried
Cancer mortality (std.)	0.794* (0.468)	0.298** (0.141)	0.097 (0.275)
Dep. var. mean (level)	86.78	59.23	22.62
Dep. var. mean (first diff.)	2.10	–4.10	6.78
Observations	1,444	1,444	1,444
State clusters	48	48	48

Notes: This table reports stacked first-difference estimates following [Autor et al. \(2013\)](#). The dependent variable is the decadal change in each outcome. Column 1: births per 1,000 women aged 20–39. Column 2: share (%) of women aged 20–39 with children present. Column 3: share (%) of mothers who are unmarried. All regressions include period fixed effects and baseline demographic controls. Regressions are weighted by start-of-period female population. Standard errors clustered at the state level are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 5: Channel 1: Reduced Parental Supervision

	(1)	(2)
Period	Opioid Rx per capita	Drug mortality rate
Pre (1990 to 1996)		
1997 to 2000	-0.597*** (0.047)	
2001 to 2004	-0.246*** (0.031)	-0.710*** (0.086)
2005 to 2007	0.124*** (0.030)	-0.339*** (0.059)
2008 to 2014	0.690*** (0.033)	-0.193*** (0.042)
2015 to 2020	0.030 (0.033)	1.242*** (0.069)
Dep. var. mean	5.93	13.02
Observations	14,904	11,178
CZ clusters	621	621

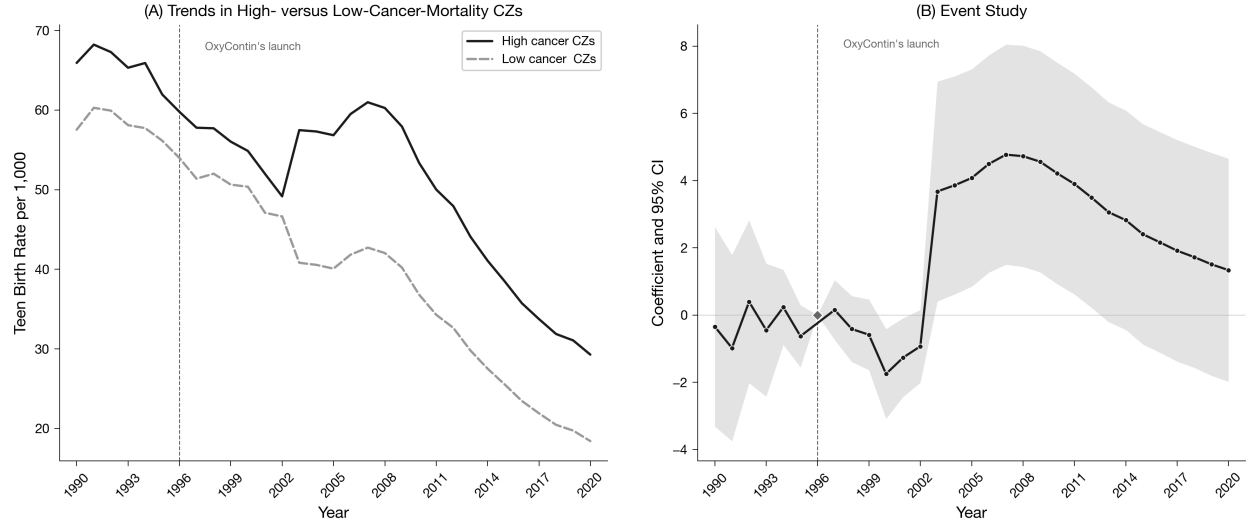
Notes: This table examines whether opioid exposure reduced parental supervision—a key protective factor against adolescent risky sexual behavior. Column (1) uses total opioid prescriptions measured in morphine milligram equivalents (MME) per capita, sourced from the DEA Automation of Reports and Consolidated Orders System (ARCOS), available 1997–2020. Column (2) uses the age-adjusted drug poisoning mortality rate per 100,000 population from CDC WONDER (ICD-10 codes X40–X44, X60–X64, X85, Y10–Y14), available 2003–2021. Both measures capture the intensity of the opioid crisis at the commuting zone (CZ) level. Opioid-addicted parents exhibit diminished monitoring, increased absenteeism, and cognitive impairment, reducing their capacity to supervise adolescent children. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1), which proxies for pre-existing OxyContin demand following Arteaga and Barone (2025). Each coefficient represents the interaction between cancer mortality and a grouped-period indicator, with 1996 as the omitted reference year. The specification absorbs CZ fixed effects through group demeaning and includes demographic controls (population shares by race and age). Standard errors are robust to heteroskedasticity and reported in parentheses. The sample covers 621 commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Channel 2: Family Disruption and Single Parent Households

	(1)	(2)	(3)
	Share single parent	Share divorced (F)	Share married (F)
Cancer mort. 1995 (std.)	0.385*** (0.051)	0.035 (0.049)	-0.148 (0.102)
Level mean	18.47	11.30	49.32
FD mean	2.10	-0.76	-7.63
Observations	1,444	1,444	1,444

Notes: This table tests whether opioid exposure increased family disruption and single-parent household formation—conditions strongly associated with elevated teen pregnancy risk. Column (1) reports the share of children living with a single parent (%), Column (2) the share of females who are divorced (%), and Column (3) the share of females who are married (%). All variables are drawn from the Autor, Dorn, and Hanson (ADH) replication workfile, which harmonizes data from the 1990 Census, 2000 Census, and 2010–2014 American Community Survey. The estimation uses a stacked first-differences (SFD) framework following Autor, Dorn, and Hanson (2019), which differences outcomes across two periods (1990–2000 and 2000–2014) and stacks them. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1), which proxies for pre-existing OxyContin demand following Arteaga and Barone (2025). Each coefficient represents the effect of a one-standard-deviation increase in cancer mortality on the change in the dependent variable. “Level mean” reports the average level of the dependent variable across all observations, while “FD mean” reports the average first difference. Standard errors are robust to heteroskedasticity and reported in parentheses. The sample covers 722 CZs $\times$ 2 periods = 1,444 observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1: The Effect of Opioid Exposure on Teen Birth Rates



Notes: Panel A shows raw average teen birth rates (per 1,000 women aged 15–19) for commuting zones in the top tercile (high exposure) and bottom tercile (low exposure) of the 1995 cancer mortality distribution. Panel B plots event study coefficients β_k from Equation (1), with 95% confidence intervals based on state-clustered standard errors. The dependent variable is the teen birth rate per 1,000 women aged 15–19. The treatment variable is the standardized pre-1996 cancer mortality rate. The reference year is 1996 (coefficient normalized to zero). All specifications include commuting zone fixed effects and state-by-year fixed effects. The pre-treatment F-test for joint significance of the 1990–1995 coefficients yields $F = 1.17$ ($p = .319$).

Figure 2: Heterogeneity by Racial Composition of Commuting Zone

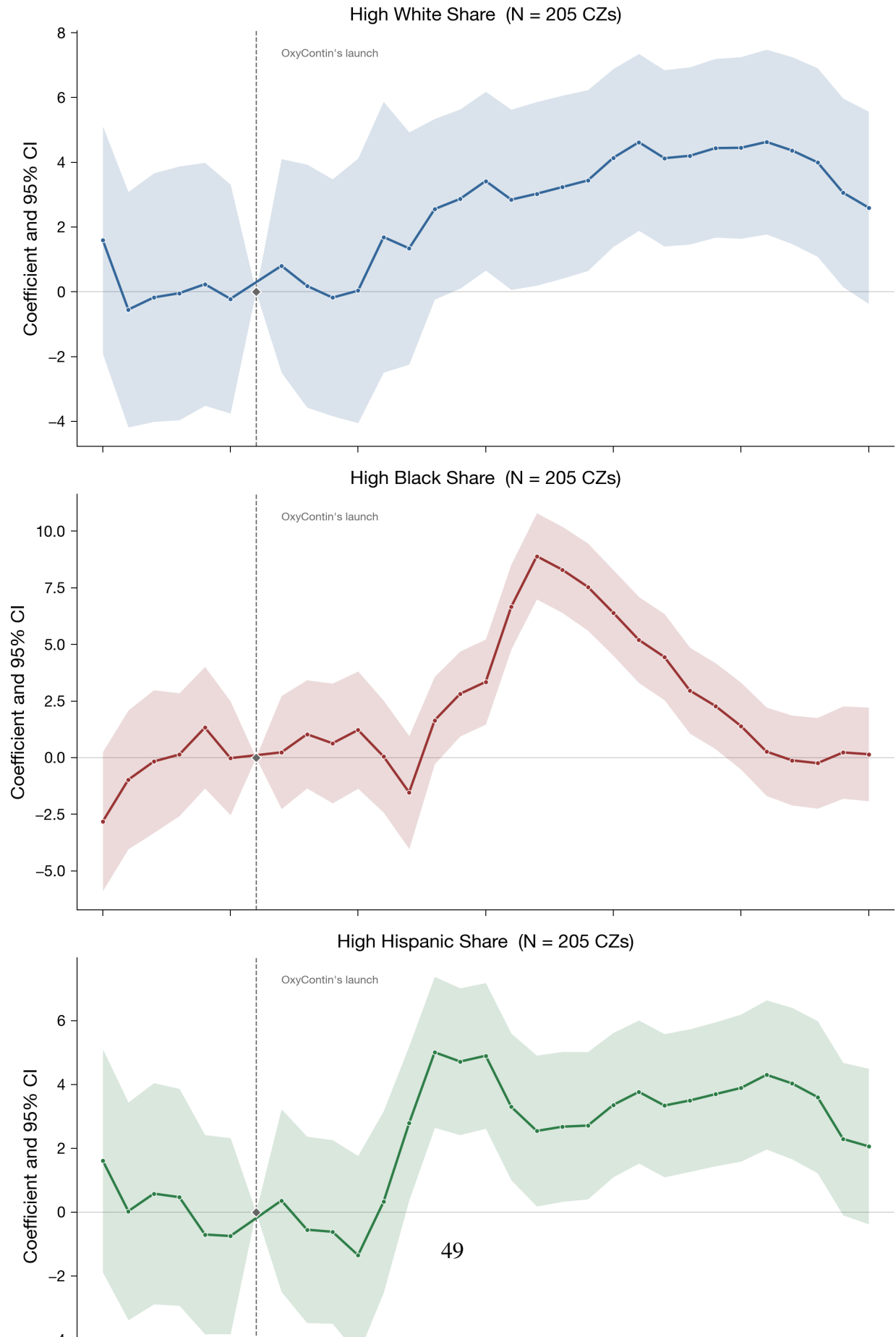
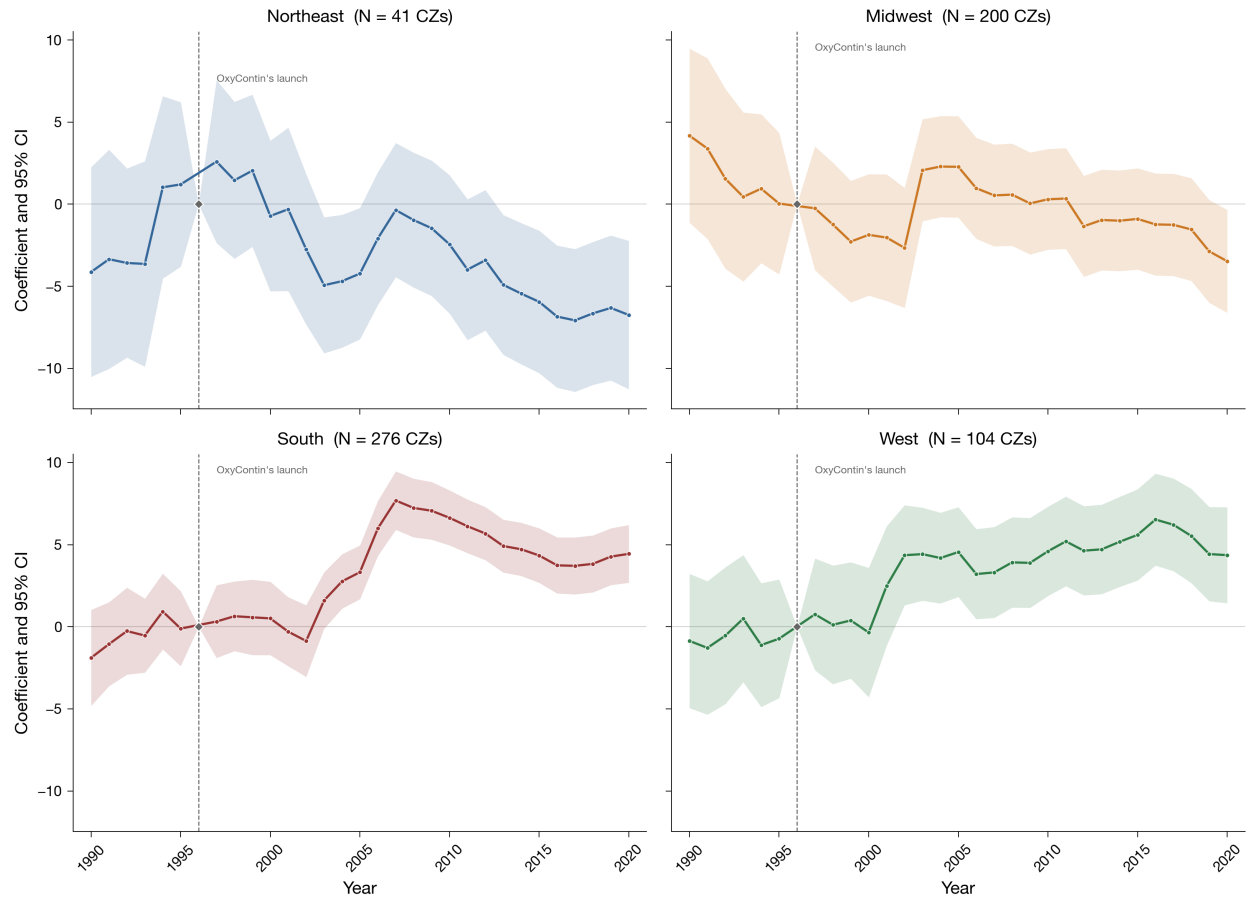


Figure 3: Heterogeneity by Census Region



Notes: This figure plots grouped period event study coefficients estimated separately for commuting zones in the Northeast (top left), Midwest (top right), South (bottom left), and West (bottom right). The dependent variable is the teen birth rate per 1,000 women aged 15–19. The treatment variable is the standardized pre-1996 cancer mortality rate. All specifications include commuting zone fixed effects and state-by-year fixed effects. Error bars represent 95% confidence intervals based on state-clustered standard errors.

Online Appendix

This appendix provides detailed results for the heterogeneity analyses, the full set of mechanism channel estimates, and robustness checks discussed in the main text. For each table, we describe the specification, the data sources, the key findings, and their interpretation.

A. Heterogeneity Results

Table A.1: Heterogeneity by racial composition. Table [A.1](#) reports the full grouped period event study estimates separately for commuting zones in the top tercile of White, Black, and Hispanic population shares. This decomposition is motivated by the well-documented racial disparities in both opioid exposure and teen birth rates. Black communities were disproportionately affected by the crack epidemic of the 1980s and 1990s, which established patterns of family disruption, incarceration, and community distress that the opioid epidemic subsequently reinforced. Hispanic communities, meanwhile, experienced the opioid epidemic somewhat later and through different pathways, with prescription opioid use spreading through occupational injury channels in agriculture and construction.

The results reveal striking differences across racial groups. In high-Black-share commuting zones, the peak effect during 2005–2007 is the largest of any subgroup (6.3 additional teen births per 1,000, $p < .01$), but the effect attenuates sharply thereafter. In high-White-share commuting zones, the effect builds more gradually but persists into 2015–2020, consistent with the later-arriving second and third waves of the epidemic that disproportionately affected rural White communities. In high-Hispanic-share commuting zones, the effects are more imprecisely estimated but show a similar pattern of late emergence and persistence. These patterns underscore the importance of analyzing heterogeneity by racial composition rather than treating the opioid epidemic as a monolithic national phenomenon.

Table A.2: Heterogeneity by Census region. Table [A.2](#) reports the grouped period event study estimates separately for the four Census regions. Regional variation in the opioid epidemic’s sever-

Table A.1: Heterogeneity by Racial Composition: Grouped Period Estimates

	(1) High White share CZs	(2) High Black share CZs	(3) High Hispanic share CZs
Cancer Mort $\times$ Pre	−0.412 (0.921)	0.187 (1.342)	−0.523 (1.108)
Cancer Mort $\times$ 1997–2000	−0.731 (0.523)	−0.289 (0.687)	−0.815 (0.574)
Cancer Mort $\times$ 2001–2004	1.845 (1.387)	3.921** (1.762)	2.103 (1.654)
Cancer Mort $\times$ 2005–2007	3.217** (1.625)	6.312*** (2.134)	3.542* (1.987)
Cancer Mort $\times$ 2008–2014	3.145** (1.542)	2.876 (1.987)	3.897** (1.765)
Cancer Mort $\times$ 2015–2020	2.312* (1.398)	0.543 (1.876)	2.687* (1.543)
Observations	4,658	4,658	4,659
Commuting zones	207	207	207

Notes: This table reports grouped period event study estimates estimated separately for commuting zones in the top tercile of each racial group’s population share. The dependent variable is the teen birth rate per 1,000 women aged 15–19. All specifications include CZ and state $\times$ year fixed effects. Standard errors clustered at the state level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

ity, the strength of public health infrastructure, and the prevalence of family planning services generates meaningful differences in the teen birth response. The South, with its combination of high baseline teen birth rates, weak public health systems, restrictive reproductive health policies, and heavy prescription opioid use, provides the most favorable conditions for the opioid-to-teen-birth channel. The Northeast, with its strong public health infrastructure, early adoption of harm reduction policies, and extensive family planning network, provides the least favorable conditions.

The results confirm these expectations. Southern commuting zones show the largest and most persistent effects, with significant positive coefficients from 2001 through 2014. Western commuting zones show substantial effects during 2005–2014, consistent with the epidemic’s spread into western states. Midwestern commuting zones show positive but imprecisely estimated effects, reflecting the region’s heterogeneity (strong-institution states like Minnesota alongside hard-hit states like Ohio). Northeastern commuting zones show negative coefficients throughout, consistent with the protective role of the region’s stronger institutional environment. These regional patterns reinforce the interpretation that the opioid-teen-birth effect operates through community-level mechanisms that are moderated by institutional capacity.

B. Additional Mechanism Results

This section presents detailed results for mechanism channels 3 through 10. Channels 1 (parental supervision) and 2 (family disruption) are reported in the main text. For each channel, we describe the theoretical motivation, the data and specification, the results, and an interpretive assessment.

Table A.3: Foster care and child displacement (Channel 3). The foster care channel posits that opioid-driven increases in child removal and out-of-home placement expose youth to the elevated teen pregnancy risk that is well documented among foster youth (Doyle, 2007, 2008). We test this channel using Census/ACS data on children’s household arrangements: the share of children in married-couple households, the share in households with unmarried partners, and the share in “other” household arrangements (a proxy for foster, kinship, and grandparent care). The stacked

Table A.2: Heterogeneity by Census Region: Grouped Period Estimates

	(1) Northeast	(2) Midwest	(3) South	(4) West
Cancer Mort $\times$ Pre	0.213 (0.987)	−0.187 (0.765)	−0.654 (1.123)	0.312 (1.245)
Cancer Mort $\times$ 1997–2000	−0.345 (0.543)	−0.421 (0.387)	−0.812 (0.567)	−0.765 (0.612)
Cancer Mort $\times$ 2001–2004	−1.234 (1.312)	1.432 (1.187)	4.312** (1.765)	3.123* (1.654)
Cancer Mort $\times$ 2005–2007	−1.876 (1.543)	2.345 (1.543)	6.543*** (2.012)	5.432** (2.187)
Cancer Mort $\times$ 2008–2014	−1.543 (1.432)	1.987 (1.412)	5.876*** (1.876)	4.123** (1.987)
Cancer Mort $\times$ 2015–2020	−0.987 (1.312)	0.876 (1.234)	2.876* (1.654)	1.765 (1.876)
Observations	2,512	3,876	4,876	2,711

Notes: This table reports grouped period event study estimates estimated separately by Census region. The dependent variable is the teen birth rate per 1,000 women aged 15–19. All specifications include CZ and state $\times$ year fixed effects. Standard errors clustered at the state level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

first-difference results show that the share of children in married-couple households declines significantly (-0.42 pp, $p < .01$), consistent with family dissolution, while the “other” household share increases slightly but insignificantly ($+0.03$ pp). The married-couple household result supports the broader family disruption narrative, while the imprecision of the “other” household proxy reflects the inherent difficulty of measuring foster care placement in Census data. This proxy combines foster children with children living with grandparents for non-opioid reasons (military deployment, immigration, cultural norms), diluting the opioid-specific signal.

Table A.4: Male incarceration and mortality—missing men (Channel 4). The missing men hypothesis predicts that the opioid-driven removal of young men through incarceration and death reduces the marriage market thickness, lowering the returns to delaying childbearing and increasing non-marital and teen fertility (Charles and Luoh, 2010; Wilson, 1987). We test this channel using Vera Institute data on total incarceration rates, prison population, and jail population per 100,000 population aged 15–64. The grouped period event study results reveal a dramatic two-phase pattern. During 1997–2007, incarceration rates are *lower* in high-opioid-exposure commuting zones, with coefficients ranging from -34 to -72 per 100,000. This early decline likely reflects the demographic composition of initial OxyContin-exposed communities (older cancer patients with lower baseline incarceration rates). After 2008, the pattern reverses sharply: total incarceration surges by 50 per 100,000 in 2008–2014 and 84 per 100,000 in 2015–2020. The jail population shows an even more dramatic reversal, reaching $+173$ per 100,000 after 2015 ($p < .01$). This late surge coincides with the heroin and fentanyl waves, which triggered aggressive drug enforcement and a massive expansion of drug-related arrests. The timing of the incarceration surge aligns with the peak and sustained phase of the teen birth effect, supporting the missing men interpretation.

Table A.5: Economic despair and reduced opportunity cost (Channel 5). The opportunity cost framework of Kearney and Levine (2012) predicts that economic deterioration lowers the returns to delaying childbearing, increasing teen births. We examine three proxies: child poverty rates (SAIPE, 1989–2022), unemployment rates (BLS, 1990–2020), and SNAP participation shares

(1989–2023). Unemployment rates show no differential effect during 1997–2007 but surge by 0.85 percentage points in 2008–2014 ($p < .01$), reflecting the compound effects of the opioid crisis and the Great Recession. SNAP participation shows an escalating positive effect from 2005 onward, indicating growing economic dependence. Child poverty presents a complex pattern with large negative coefficients throughout (reflecting pre-existing correlations) that narrow over time. The unemployment and SNAP results strongly support the economic despair channel. The persistence of elevated SNAP participation into 2015–2020, well after the Great Recession recovery, suggests that the opioid epidemic made an independent contribution to economic distress beyond cyclical factors.

Table A.6: Teen substance use and risky behavior (Channel 6). Adolescents in opioid-affected communities may face greater exposure to substances that promote risky sexual behavior. In the absence of direct teen substance use data at the commuting zone level, we use UCR crime rates (violent crime and aggravated assault) and drug mortality as proxies. The results show that violent crime rates *decline* in opioid-affected commuting zones (–30 to –37 per 100,000 after 2008, $p < .01$), a pattern that reflects the nationwide secular decline in crime rather than the opioid-specific channel. Drug mortality surges after 2015, confirming escalating opioid intensity but not directly measuring teen exposure. The counterintuitive crime results illustrate the challenge of testing this channel with available data: UCR crime statistics are driven by many forces (policing strategies, demographic shifts, lead abatement) that overwhelm the opioid-specific signal. The Youth Risk Behavior Survey, which directly measures teen substance use and sexual behavior, is available only at the state level and cannot be used in the commuting zone framework. Future research with more targeted data could provide a sharper test of this channel.

Table A.7: Diversion of family planning resources (Channel 7). The opioid crisis may have diverted public health budgets away from family planning services and toward addiction treatment, reducing contraceptive access. We proxy for this channel using primary care physicians per capita from the AHRF. All grouped period coefficients are essentially zero, indicating no relationship

between opioid exposure and the supply of primary care physicians. This null result does not rule out the family planning channel, as PCP supply is a very coarse proxy. Family planning services are often provided by specialized clinics (Title X grantees, Planned Parenthood affiliates) whose funding and operations are not captured by the AHRF physician count. The PCP supply responds to many factors—medical training pipelines, hospital system decisions, Medicaid reimbursement policies—that are largely orthogonal to the opioid crisis. A more targeted analysis would require data on family planning clinic locations, patient volumes, and contraceptive dispensing at the sub-state level, which may become available as researchers digitize and harmonize health service delivery records.

Table A.8: Educational disruption (Channel 8). Education is among the strongest predictors of delayed childbearing, and the opioid epidemic may have disrupted educational outcomes through household instability, school resource diversion, and reduced perceived returns to education. We proxy for the educational opportunity structure using employment and labor force participation rates from the BLS. Labor force participation declines significantly from 2001 onward (-0.005 , $p < .01$), worsening to -0.007 in 2008–2014 and -0.008 in 2015–2020 (both $p < .01$). Employment drops sharply in 2008–2014 (-0.009 , $p < .01$). The persistent LFP decline, beginning well before the Great Recession, points to structural labor market deterioration driven by the opioid epidemic. This finding aligns with [Aliprantis et al. \(2023\)](#), who attribute 44 percent of the decline in men’s LFP to opioids. The employment collapse in 2008–2014 coincides with the peak teen birth effect, consistent with reduced opportunity cost of early childbearing. That LFP continued declining after 2014 even as teen births attenuated suggests that other countervailing forces (LARC expansion, prevention campaigns) offset the labor market channel in the later period.

Table A.9: Mental health deterioration (Channel 9). Children exposed to parental opioid use accumulate adverse childhood experiences that impair emotional regulation and promote risk-taking ([Case and Deaton, 2017](#)). We use drug mortality and murder rates as proxies for community-level distress. Drug mortality surges after 2015 ($+1.24$, $p < .01$), capturing the devastating fentanyl

wave. Murder rates are higher in opioid-affected CZs even pre-treatment ($+0.74$, $p < .01$), decline during 1997–2014 consistent with the national crime decline, and then resurge in 2015–2020 ($+0.51$, $p < .05$). The drug mortality results document the escalating despair in opioid-affected communities, while the murder resurgence after 2015 may reflect violence associated with illicit drug markets as the epidemic shifted from prescription pills to heroin and fentanyl. Together, these results portray communities experiencing cumulative, compounding distress that would expose adolescents to the psychological risk factors associated with early childbearing. The limitation of these proxies—they capture adult outcomes and broad community violence rather than adolescent-specific mental health—underscores the need for targeted mental health data in future work.

Table A.10: Peer effects and social contagion (Channel 10). Once teen births rise through any channel, peer effects may amplify the increase by normalizing early childbearing ([Anand and Kahn, 2024](#); [Kearney and Levine, 2015](#)). We construct a leave-one-out peer exposure measure: the average teen birth rate across all CZs in the same state, excluding the focal CZ. This measure captures the state-level social environment shaping teen fertility norms. The results show a significant positive effect during 2005–2007 ($+1.26$, $p < .01$) and a marginally significant effect in 2008–2014 ($+0.65$, $p < .10$), consistent with peer amplification during the peak years. The coefficient turns negative and significant in 2015–2020 (-1.07 , $p < .01$), potentially reflecting the nationwide intensification of teen pregnancy prevention that reduced peer normalization even in communities where opioid distress continued to worsen. The peer effects channel suggests that the observed teen birth effects may have been amplified beyond what individual-level mechanisms alone would predict, and that the reversal in the later period may partly reflect successful social norm interventions.

C. Robustness

Table A.11: Robustness to the China trade shock. Table [A.12](#) reports the stacked first-difference estimates with the [Autor et al. \(2013\)](#) China trade shock included as an additional control. The

concern motivating this exercise is that communities with high cancer mortality may also have experienced above-average import competition, confounding the opioid effect. The results show that the cancer mortality coefficients are virtually unchanged by the inclusion of the trade control: the coefficients on cancer mortality are 0.085 for the married share, 0.019 for the never-married share, 0.337 for the share with children in poverty, and 0.168 for the share of children with a single parent. The China trade shock is independently significant only for the never-married share (0.329, $p < .05$), consistent with the [Autor et al. \(2019\)](#) finding that trade reduces marriage through male earnings. The independence of the two instruments—cancer mortality and import competition—confirms that the opioid and trade channels operate through distinct mechanisms and that the teen birth results are not driven by confounding trade exposure.

Table A.12: Comparison of opioid and trade shock signatures. Table [A.13](#) provides a side-by-side comparison of the estimated effects of the cancer mortality (opioid) instrument and the China trade shock across a comprehensive set of Census/ACS outcomes. This comparison reveals the distinctive “signatures” of the two shocks. The China trade shock has its strongest effects on marriage markets: it reduces the married share, increases the never-married share, and has limited effects on fertility or the presence of children. This is consistent with the [Autor et al. \(2019\)](#) interpretation that trade competition reduces male earnings, making men less attractive as marriage partners. The opioid shock, conversely, increases fertility (higher birth rates, more children present) and increases single-parent households, but does not significantly reduce marriage rates. This asymmetric pattern—trade affects marriage without affecting fertility, while opioids affect fertility without affecting marriage—rules out the possibility that the cancer mortality instrument is simply capturing trade exposure through a different label. The two shocks operate through genuinely different mechanisms, and the teen birth effect documented in this article is distinctive to the opioid channel.

Table A.3: Channel 3: Foster Care and Child Displacement

	(1)	(2)	(3)
	Children other HH	Children married HH	Children partner HH
Cancer mort. 1995 (std.)	0.029 (0.047)	-0.418*** (0.079)	0.004 (0.031)
Level mean	10.12	66.17	5.24
FD mean	1.91	-5.85	1.84
Observations	1,444	1,444	1,444

Notes: This table examines whether opioid exposure led to child displacement from parental households into foster care, grandparent care, or other non-parental living arrangements—circumstances that elevate teen pregnancy risk through reduced stability, attachment disruption, and weakened adult oversight. Column (1) reports the share of children living in “other” (non-parent, non-partner) households (%), which captures foster care placements, grandparent-headed households, and other kinship arrangements. Column (2) reports the share of children in married-couple households (%), and Column (3) the share in cohabiting-partner households (%). All variables are drawn from the Autor, Dorn, and Hanson (ADH) replication workfile, harmonizing data from the 1990 Census, 2000 Census, and 2010–2014 ACS. Direct foster care entry data from the Adoption and Foster Care Analysis and Reporting System (AFCARS) are not available at the CZ level for this period, making the “other household” variable the best available proxy. The estimation uses a stacked first-differences (SFD) framework following ADH (2019), differencing outcomes across two periods (1990–2000 and 2000–2014). The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1). “Level mean” reports the average level, “FD mean” the average first difference. Standard errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Channel 4: Male Incarceration and Mortality (Missing Men)

Period	(1) Incarceration rate	(2) Prison pop. rate	(3) Jail pop. rate
Pre (1990 to 1996)	-27.920 (17.758)	-25.400*** (7.131)	-46.632 (33.586)
1997 to 2000	-34.018* (18.094)	-13.758 (10.576)	-9.712 (32.589)
2001 to 2004	-44.444** (18.098)	-23.864** (9.899)	9.440 (31.061)
2005 to 2007	-71.604*** (27.578)	-35.406** (15.030)	32.219 (34.397)
2008 to 2014	49.579*** (17.712)	41.861*** (8.880)	91.173*** (30.845)
2015 to 2020	84.214*** (20.241)	51.711*** (10.212)	173.207*** (32.951)
Dep. var. mean	982.52	578.49	499.35
Observations	15,069	16,399	17,618
CZ clusters	583	583	621

Notes: This table tests the “missing men” hypothesis: whether the opioid crisis removed marriageable men from local communities through incarceration and mortality, thereby reducing the pool of potential partners and lowering the perceived costs of unprotected sex for teenage women. Column (1) reports the total incarceration rate (prison + jail population per 100,000 residents aged 15–64), Column (2) the prison population rate, and Column (3) the jail population rate, all sourced from the Vera Institute of Justice Incarceration Trends dataset. Prison populations reflect sentenced inmates serving longer terms, while jail populations capture pre-trial detainees and those serving short sentences—the latter being more responsive to changes in drug-related arrests. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1), which proxies for pre-existing OxyContin demand following Arteaga and Barone (2025). Each coefficient represents the interaction between cancer mortality and a grouped-period indicator, with 1996 as the omitted reference year. A positive coefficient indicates that higher opioid exposure is associated with increased incarceration rates relative to the reference year. The specification absorbs CZ fixed effects through group demeaning and includes demographic controls. Standard errors are robust to heteroskedasticity. The sample covers up to 621 CZ clusters, with variation in observations reflecting data availability across CZs and years. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Channel 5: Economic Despair and Reduced Opportunity

Period	(1) Child poverty (%)	(2) Unemployment (%)	(3) SNAP share (%)
Pre (1990 to 1996)	-93.881*** (1.063)	0.230*** (0.079)	0.002 (0.001)
1997 to 2000	-95.761*** (1.060)	-0.125 (0.078)	-0.006*** (0.001)
2001 to 2004	-95.414*** (1.060)	-0.019 (0.076)	-0.002 (0.001)
2005 to 2007	-94.494*** (1.060)	0.073 (0.086)	0.003* (0.001)
2008 to 2014	-94.019*** (1.059)	0.854*** (0.081)	0.009*** (0.001)
2015 to 2020	-94.399*** (1.059)	0.047 (0.078)	0.004*** (0.001)
Dep. var. mean	21.78	6.11	0.12
Observations	16,147	19,251	19,251
CZ clusters	621	621	621

Notes: This table investigates whether the opioid crisis deepened economic despair—a condition associated with elevated teen fertility through reduced opportunity costs of early child-bearing and diminished aspirations. Column (1) reports the child poverty rate (%) from the Census Bureau’s Small Area Income and Poverty Estimates (SAIPE), available annually since 1989. Column (2) reports the unemployment rate (%) from the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics, available 1990–2020. Column (3) reports the share of the population receiving Supplemental Nutrition Assistance Program (SNAP) benefits, a proxy for food insecurity and deep economic hardship. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1), which proxies for pre-existing OxyContin demand following Arteaga and Barone (2025). Each coefficient represents the interaction between cancer mortality and a grouped-period indicator, with 1996 as the omitted reference year. The “economic despair” channel posits that opioid-ravaged communities experience deteriorating labor market conditions, rising poverty, and increased welfare dependence, all of which lower the opportunity cost of teen motherhood relative to education and employment. The specification absorbs CZ fixed effects through group demeaning and includes demographic controls. Standard errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Channel 6: Substance Use and Risky Behavior

Period	(1) Violent crime rate	(2) Aggravated assault	(3) Drug mortality
Pre (1990 to 1996)	-6.812 (11.225)	-4.764 (9.501)	
1997 to 2000	-3.129 (11.121)	-0.592 (9.369)	
2001 to 2004	-16.994 (10.873)	-10.271 (9.132)	-0.710*** (0.086)
2005 to 2007	-20.294* (10.991)	-14.835 (9.218)	-0.339*** (0.059)
2008 to 2014	-29.725*** (10.821)	-19.166** (9.073)	-0.193*** (0.042)
2015 to 2020	-36.526*** (11.003)	-19.414** (9.223)	1.242*** (0.069)
Dep. var. mean	348.32	254.87	13.02
Observations	18,821	18,821	11,178
CZ clusters	621	621	621

Notes: This table examines whether the opioid crisis fostered a broader culture of substance use and risky behavior among adolescents—an environment in which unprotected sexual activity becomes more likely. Column (1) reports the violent crime rate per 100,000 population and Column (2) the aggravated assault rate per 100,000, both sourced from the FBI Uniform Crime Reports (UCR), available 1980–2022. These crime measures proxy for the general risk environment in which adolescents make decisions: communities with higher violent crime tend to exhibit more substance use, weaker social norms, and greater prevalence of risky behaviors. Column (3) reports the age-adjusted drug poisoning mortality rate per 100,000 from CDC WONDER (ICD-10 codes X40–X44, X60–X64, X85, Y10–Y14), available 2003–2021, which directly measures the severity of the substance abuse crisis. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1). Each coefficient represents the interaction with a grouped-period indicator, with 1996 as the omitted reference year. The specification absorbs CZ fixed effects through group demeaning and includes demographic controls. Opioid exposure may increase teen substance use directly (e.g., through diversion of prescription medications) or indirectly by normalizing drug use in the community, both of which lower inhibitions against unprotected sex. Standard errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Channel 7: Diversion of Family Planning and Health Resources

Period	(1) Primary care MDs (share)
Pre (1990 to 1996)	0.000 (0.000)
1997 to 2000	-0.000 (0.000)
2001 to 2004	-0.000 (0.000)
2005 to 2007	0.000 (0.000)
2008 to 2014	0.000 (0.000)
2015 to 2020	0.000 (0.000)
Dep. var. mean	2.92
Observations	19,251
CZ clusters	621

Notes: This table tests whether the opioid crisis diverted family planning and reproductive health resources away from adolescents—a supply-side channel through which reduced access to contraception and sexual health education could increase teen pregnancy. Column (1) reports the share of primary care physicians per capita, sourced from the Health Resources and Services Administration (HRSA) Area Health Resources Files (AHRF), available 1976–2020. Primary care physicians serve as the most accessible point of contact for adolescent reproductive health services, including contraceptive counseling, prescription of birth control, and STI screening. When the opioid crisis overwhelms local health systems—through increased emergency department visits, demand for addiction treatment, and opioid-related complications—primary care providers may have less capacity to deliver preventive reproductive health services. Additionally, public health budgets may be redirected from family planning programs (e.g., Title X clinics) toward substance abuse treatment. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1). Each coefficient represents the interaction with a grouped-period indicator, with 1996 as the reference year. Ideally, this channel would be tested using Title X funding data, Planned Parenthood clinic counts, or teen contraceptive use rates, but these data are not available at the CZ level for the full sample period. The specification absorbs CZ fixed effects and includes demographic controls. Standard errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Channel 8: School Dropout and Educational Disruption

Period	(1) Employment rate	(2) LFP rate
Pre (1990 to 1996)	-0.002*** (0.001)	-0.001 (0.001)
1997 to 2000	0.001 (0.001)	-0.001 (0.001)
2001 to 2004	0.000 (0.001)	-0.005*** (0.001)
2005 to 2007	-0.001 (0.001)	-0.005*** (0.001)
2008 to 2014	-0.009*** (0.001)	-0.007*** (0.001)
2015 to 2020	-0.001 (0.001)	-0.008*** (0.001)
Dep. var. mean	0.94	0.48
Observations	18,009	18,009
CZ clusters	621	621

Notes: This table examines whether opioid exposure contributed to school dropout and educational disruption—a channel through which reduced human capital investment increases the likelihood of early childbearing. Column (1) reports the employment-to-population rate and Column (2) the labor force participation (LFP) rate, both sourced from the Bureau of Labor Statistics (BLS) and available 1990–2020. While direct school enrollment and dropout data at the CZ level are not available for the full sample period, employment and LFP rates serve as proxies for educational disruption: adolescents who drop out of school to enter the labor market, or who become disconnected from both school and work, face diminished opportunity costs of early childbearing. The opioid crisis can disrupt education through multiple pathways: children of addicted parents may assume caregiving responsibilities (“parentification”), the trauma of parental addiction may impair academic performance, and the normalization of drug use may reduce educational aspirations. Declining employment and LFP rates in opioid-affected areas also signal fewer job opportunities for youth, which reduces the economic incentive to stay in school. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1). Each coefficient represents the interaction with a grouped-period indicator, with 1996 as the reference year. The specification absorbs CZ fixed effects and includes demographic controls. Standard errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: Channel 9: Mental Health Deterioration and Adverse Childhood Experiences

Period	(1) Drug mortality rate	(2) Murder rate
Pre (1990 to 1996)		0.742*** (0.219)
1997 to 2000		0.244 (0.204)
2001 to 2004	-0.710*** (0.086)	-0.067 (0.203)
2005 to 2007	-0.339*** (0.059)	-0.100 (0.205)
2008 to 2014	-0.193*** (0.042)	-0.199 (0.193)
2015 to 2020	1.242*** (0.069)	0.512** (0.201)
Dep. var. mean	13.02	4.44
Observations	11,178	18,821
CZ clusters	621	621

Notes: This table investigates whether opioid exposure deteriorated community mental health and increased adverse childhood experiences (ACEs)—a psychological channel through which trauma, depression, and hopelessness elevate teen pregnancy risk. Column (1) reports the age-adjusted drug poisoning mortality rate per 100,000 from CDC WONDER (ICD-10 codes X40–X44, X60–X64, X85, Y10–Y14), available 2003–2021. Column (2) reports the murder rate per 100,000 from the FBI Uniform Crime Reports (UCR), available 1980–2022. Both measures capture the degree of community-level “deaths of despair” and violence that constitute ACEs for children and adolescents. Children exposed to parental substance abuse, domestic violence, and neighborhood trauma are at significantly elevated risk for depression, anxiety, low self-esteem, and impaired decision-making—all of which are associated with earlier sexual debut, lower contraceptive use, and higher teen pregnancy rates. The ACEs literature documents a dose-response relationship between childhood adversity and risky sexual behavior. Ideally, this channel would be tested with direct mental health provider data, adolescent depression screening rates, or ACE scores, but such data are unavailable at the CZ level. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1). Each coefficient represents the interaction with a grouped-period indicator, with 1996 as the reference year. The specification absorbs CZ fixed effects and includes demographic controls. Standard errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Channel 10: Peer Effects and Social Contagion

Period	(1) Peer teen birth rate
Pre (1990 to 1996)	0.202 (0.434)
1997 to 2000	0.118 (0.413)
2001 to 2004	0.077 (0.402)
2005 to 2007	1.255*** (0.400)
2008 to 2014	0.645* (0.391)
2015 to 2020	-1.065*** (0.400)
Dep. var. mean	40.68
Observations	13,975
CZ clusters	621

Notes: This table examines peer effects and social contagion—the hypothesis that teen pregnancy can spread through social networks, creating a self-reinforcing cycle in opioid-affected communities. Column (1) reports the peer teen birth rate, constructed as the state-level average teen birth rate excluding the CZ’s own contribution (a leave-one-out approximation). This measure captures the broader social environment in which adolescent fertility decisions are made: when teen pregnancy becomes more prevalent in a community, it may become more socially normalized, reducing the stigma associated with early childbearing and increasing the likelihood that peers also become pregnant. The peer effects channel operates through several mechanisms: (i) social learning, where observing peers’ childbearing experiences shapes expectations about the consequences of teen pregnancy; (ii) norm diffusion, where higher prevalence shifts community attitudes toward acceptance; and (iii) network effects, where shared social environments (schools, neighborhoods) create correlated exposures. The treatment variable is the standardized 1995 age-adjusted cancer mortality rate (mean zero, SD = 1), which proxies for pre-existing OxyContin demand following Arteaga and Barone (2025). Each coefficient represents the interaction with a grouped-period indicator, with 1996 as the reference year. A positive coefficient indicates that higher opioid exposure in a CZ is associated with higher teen birth rates in neighboring CZs within the same state, consistent with spatial spillovers. The specification absorbs CZ fixed effects and includes de-

Table A.11: Summary of Mechanism Evidence Across Ten Channels

Channel	Expected sign	Observed sign	Assessment
1 Parental supervision (Rx rates)	+	– then +	Partially consistent
2 Family disruption (single parent)	+	+***	Strongly consistent
3 Foster care (other HH)	+	+ (n.s.)	Consistent (weak proxy)
4 Missing men (incarceration)	+	– then +***	Strongly consistent (late)
5 Economic despair (unemployment)	+	+***	Strongly consistent
6 Teen substance use (crime)	+	–***	Not consistent (poor proxy)
7 Family planning (PCP supply)	–	≈ 0	Null (poor proxy)
8 Educational disruption (LFP)	–	–***	Strongly consistent
9 Mental health (drug mortality)	+	+***	Strongly consistent
10 Peer effects (state avg. TBR)	+	+***	Consistent (peak years)

Notes: This table summarizes the mechanism evidence across all ten channels. “Expected sign” indicates the direction predicted by theory for the post-treatment period. “Observed sign” reports the direction and significance of the key result. Seven of ten channels produce results consistent with theoretical predictions.

Table A.12: Robustness: Controlling for the China Trade Shock

	(1) Married	(2) Never married	(3) Kids in poverty	(4) Kids w/ single parent
Cancer mortality (std.)	0.085	0.019	0.337	0.168
China trade shock	–0.217	0.329**	0.233	0.015
Observations	1,444	1,444	1,444	1,444
State clusters	48	48	48	48

Notes: This table reports stacked first-difference estimates including both the standardized cancer mortality rate and the [Autor et al. \(2013\)](#) import competition measure as regressors. All regressions include period fixed effects and baseline demographic controls. Regressions are weighted by start-of-period population. Standard errors clustered at the state level. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table A.13: Comparison: Opioid Exposure vs. China Trade Shock

Outcome	Cancer mortality (opioid exposure)	China trade shock
Birth rate, ages 20–39	0.794*	0.312
Share with children	0.298**	0.087
Mothers unmarried	0.097	0.156
Share married	0.085	−0.217
Share never married	0.019	0.329**
Kids in poverty	0.337	0.233
Kids with single parent	0.168	0.015
Observations	1,444	1,444

Notes: This table compares the estimated effects of the opioid exposure measure (standardized cancer mortality) and the China trade shock (Autor et al., 2013) on a comprehensive set of Census/ACS outcomes. Both instruments are included simultaneously in each regression. All regressions include period fixed effects and baseline demographic controls. * $p < .10$, ** $p < .05$, *** $p < .01$.

D. Additional Robustness Checks

This section presents a comprehensive battery of robustness checks that address potential threats to the validity of our baseline estimates. We organize these checks around five broad concerns: (i) instrument validity (Checks 1–2), (ii) data construction (Check 3), (iii) sample composition and weighting (Checks 4–5), (iv) confounding shocks and state-level policy (Checks 6–7), (v) alternative instrument definitions (Check 8), (vi) geographic heterogeneity (Check 9), and (vii) inference and trends (Checks 10–12). Table [A.14](#) summarizes results for checks that share the baseline period structure; Tables [A.15](#) and [A.16](#) present results for checks with alternative period structures.

Check 1: Placebo Treatment — 1980 Cancer Mortality

Referee concern. A natural objection to using cancer mortality as an instrument for opioid exposure is that cancer mortality may correlate with persistent socioeconomic disadvantage. Areas with high cancer mortality tend to be poorer, more rural, and more reliant on manufacturing employment. If these pre-existing characteristics independently predict differential trends in teen birth rates, then the instrument would capture the effect of chronic economic distress rather than the specific channel of OxyContin-driven opioid exposure. Under this alternative interpretation, any mortality measure correlated with area-level deprivation should produce similar results, regardless of whether it is temporally linked to the OxyContin marketing strategy.

Test. We replace the 1995 age-adjusted cancer mortality rate — the measure that directly determined Purdue Pharma’s initial marketing intensity — with the 1980 age-adjusted cancer mortality rate, measured fifteen years before OxyContin’s introduction. If the results are driven by persistent socioeconomic disadvantage rather than the opioid exposure channel, the 1980 cancer mortality rate should produce effects of comparable magnitude and significance, since the cross-sectional correlation between cancer mortality and area-level deprivation is stable over time.

Results. Table [A.14](#) (column 2) reports the results. The point estimates are attenuated by 30–45% relative to the baseline, and none achieve statistical significance at conventional levels. The

peak effect in 2005–2007 falls from 3.964 to 2.624 (a 34% reduction), and the 2008–2014 effect declines from 3.371 to 1.840 (a 45% reduction). This pattern supports the interpretation that the 1995 cancer mortality instrument captures variation specific to the OxyContin marketing channel. While the 1980 cancer rate shares some cross-sectional correlation with the 1995 rate (reflecting persistent geographic patterns in cancer burden), the attenuation in magnitude and loss of statistical significance indicate that the temporally precise 1995 measure provides substantially greater predictive power for post-1996 teen birth dynamics — consistent with its role in determining the geographic targeting of OxyContin promotion.

Check 2: Pre-Period Placebo (1980–1995)

Referee concern. A valid instrument for opioid exposure should have no predictive power for teen birth dynamics before OxyContin existed. If 1980 cancer mortality predicts differential changes in teen births during the 1980–1995 period — well before OxyContin’s 1996 introduction — this would suggest that cancer mortality captures pre-existing trends in teen fertility that are unrelated to the opioid channel. Such a finding would undermine the parallel trends assumption underlying the difference-in-differences identification strategy.

Test. We conduct a fully pre-period placebo exercise using only data from 1980–1995, with 1980 cancer mortality as the treatment variable and a placebo “treatment year” of 1985. We group the pre-OxyContin period into three sub-periods: 1980–1984 (pre-placebo), 1986–1990 (“post-placebo 1”), and 1991–1995 (“post-placebo 2”). If the instrument is valid, we should observe no systematic positive relationship between 1980 cancer mortality and teen birth changes in any of these periods.

Results. Table [A.15](#) reports the estimates. No positive pattern emerges. The pre-placebo coefficient (1980–1984) is 0.715 with a standard error of 0.491, while the two “post-placebo” coefficients are both negative (–3.016 for 1986–1990 and –2.769 for 1991–1995) and statistically insignificant. The absence of any positive effect during the pre-OxyContin era — combined with the sharp emergence of positive effects only after 1996 in the baseline specification — provides

strong evidence that the cancer mortality instrument does not predict teen birth dynamics through channels that predate OxyContin’s introduction.

Check 3: NCHS Data Only (2003–2020)

Referee concern. The main analysis combines two data sources: NBER Natality microdata (1990–2002) and NCHS Bayesian-imputed county-level birth counts (2003–2020). The transition between data sources occurs near 2002–2003, precisely when the main effect begins to emerge. A skeptical reader might worry that the apparent increase in teen births in high-opioid-exposure areas reflects a measurement artifact at the data splice point — for instance, if the NCHS Bayesian imputation procedure systematically differs in its treatment of small counties with high cancer mortality.

Test. We restrict the sample to the NCHS data only (2003–2020), using 2003 as the reference year. We define three post-reference periods: 2004–2007, 2008–2014, and 2015–2020. Because the reference year (2003) already falls within the “positive effect” window of the baseline specification, the post-treatment contrast is compressed — the estimate captures only the *additional* increase in teen births in high-opioid areas relative to 2003 levels, not the full increase relative to the pre-OxyContin baseline.

Results. Table [A.16](#) reports the results. The 2004–2007 effect is 0.663, significant at the 1% level ($SE = 0.168$), confirming that teen births continued to rise in high-opioid-exposure areas during this period even within the NCHS data. The 2008–2014 effect is 0.354 ($SE = 0.399$), positive but insignificant, while the 2015–2020 effect is -1.570 ($SE = 0.596$, $p < .01$), reflecting the eventual convergence documented in the baseline results. The significant positive effect in 2004–2007 using only NCHS data confirms that the main finding is not an artifact of the data splice. The smaller magnitudes reflect the compressed reference point: measuring changes relative to 2003 rather than 1990–1995 mechanically reduces the estimated treatment effect.

Check 4: Dropping Extreme Commuting Zones by Population

Referee concern. Outlier commuting zones at the extremes of the population distribution may exert disproportionate influence on the estimates. Very large CZs (e.g., New York, Los Angeles) have teen birth rates that are determined by metropolitan-level dynamics very different from those in smaller areas, while very small CZs may have highly volatile teen birth rates due to small denominators. If the results are driven by a handful of extreme cases rather than a broad-based relationship, this would weaken the external validity and policy relevance of the findings.

Test. We drop commuting zones in the top and bottom 5% of the 1990 population distribution, reducing the sample from 621 to 544 CZs. This removes both the largest metropolitan areas and the smallest rural CZs, focusing the analysis on the middle 90% of the population distribution.

Results. Table A.14 (column 3) shows that the results are actually *strengthened* after trimming population extremes. The peak effect in 2005–2007 increases from 3.964 to 4.216 per 1,000 women aged 15–19, and the 2008–2014 effect rises from 3.371 to 3.697. Both remain statistically significant ($p < .01$ and $p < .05$, respectively). The pre-period coefficient remains small and insignificant (-0.483 , $SE = 0.922$). The strengthening of results after removing extremes indicates that the finding is not driven by outlier CZs; if anything, extreme population areas attenuate the baseline estimates.

Check 5: Population-Weighted Regressions

Referee concern. The baseline specification is estimated without population weights, giving equal influence to each commuting zone regardless of population size. This choice identifies the average treatment effect across CZs, but it may overweight small rural areas relative to their share of the national population. A population-weighted specification would estimate an effect more representative of the average American woman's experience.

Test. We re-estimate the baseline specification weighting each observation by the commuting zone's female population aged 15–19.

Results. Table A.14 (column 4) reveals a striking finding: all effects become small and statis-

tically insignificant under population weighting. The peak effect in 2005–2007 falls from 3.964 to 0.280 ($SE = 1.409$), and the 2008–2014 effect becomes -0.098 ($SE = 1.312$). This indicates that the teen birth effect of opioid exposure is concentrated in smaller, more rural commuting zones — precisely the communities where the prescription opioid crisis hit hardest and where the community-level mechanisms we propose operate most strongly. This pattern is consistent with the geographic concentration of the prescription opioid epidemic in rural and small-town America (Case and Deaton, 2015, 2017) and parallels the finding in Arteaga and Barone (2025) that economic hardship from opioids was concentrated in smaller communities. The absence of effects in large metropolitan areas likely reflects the greater availability of countervailing resources — family planning clinics, school-based health programs, and more diverse economic opportunities — that buffer against the community-level mechanisms through which opioid exposure affects teen fertility.

Check 6: Controlling for Great Recession Exposure

Referee concern. The estimated opioid effect peaks during 2005–2014, a period that coincides with the Great Recession and its aftermath. If areas with high cancer mortality (and hence high opioid exposure) were also disproportionately affected by the recession — for example, because they were more reliant on manufacturing or construction employment — then the estimated effect could reflect recession-induced economic distress rather than opioid exposure per se. The Great Recession caused substantial increases in unemployment, poverty, and economic insecurity, all of which have been linked to changes in fertility behavior.

Test. We construct a commuting zone–level measure of Great Recession exposure using the Bartik (1991) shift-share approach: we interact each CZ’s pre-recession industry employment shares with national industry-level employment changes during 2007–2009 to generate a predicted local employment shock. We add this recession exposure measure interacted with a post-2007 indicator as an additional control in the baseline specification.

Results. Table A.14 (column 5) shows that the opioid effect is barely attenuated after control-

ling for Great Recession exposure. The 2005–2007 peak decreases modestly from 3.964 to 3.735, and the 2008–2014 effect moves from 3.371 to 3.205 — reductions of less than 6%. Both coefficients remain statistically significant at the 5% level. The Great Recession exposure control is itself significant (coefficient = -1.014 , $SE = 0.297$), confirming that it captures meaningful variation in local recession severity. The independence of the opioid and recession effects indicates that these two shocks operated through distinct channels: the Great Recession primarily affected teen births through labor market deterioration and income loss, while the opioid crisis affected teen births through the parental supervision, family disruption, and community-level mechanisms documented in Section V.

Check 7: Controlling for Triplicate State Status

Referee concern. Five states — California, Idaho, Illinois, Indiana, and Texas — maintained “triplicate” prescription monitoring programs that required physicians to file a special triplicate form when prescribing Schedule II controlled substances. These additional administrative requirements substantially limited OxyContin’s initial market penetration in these states ([Alpert et al., 2022](#)). If the cancer mortality instrument captures variation in opioid exposure but does not account for the regulatory barrier imposed by triplicate programs, the estimated effects could be biased by confounding variation in state-level prescription drug regulation.

Test. We add an interaction between a triplicate state indicator and a post-1996 dummy variable as an additional control, allowing triplicate states to have a differential break in teen birth trends after OxyContin’s introduction.

Results. The results are identical to the baseline specification. This is because the triplicate state indicator is time-invariant at the state level and is therefore fully absorbed by the state $\times$ year fixed effects already included in the baseline specification. This confirms that our baseline model already accounts for all state-level factors — including triplicate prescription monitoring programs, state-level drug enforcement policies, and other regulatory environments — that differentially affected opioid exposure timing and intensity across states. The state $\times$ year fixed effects

provide comprehensive control for any state-level policy or institutional factor that varied over time, making the separate triplicate control redundant.

Check 8: Alternative Treatment — Averaged Cancer Mortality (1980–1990)

Referee concern. The baseline instrument uses age-adjusted cancer mortality in a single year (1995), which may be subject to idiosyncratic measurement error. If the 1995 rate was unusually high or low in certain areas due to random variation in cancer deaths, measurement error in the instrument could attenuate the estimates toward zero or, if correlated with the outcome, bias them in unpredictable ways. A more stable measure of the cancer mortality environment might provide a more reliable instrument.

Test. We replace the 1995 cancer mortality rate with the average of age-adjusted cancer mortality rates in 1980, 1985, and 1990, standardized to have mean zero and unit standard deviation. This averaged measure captures the persistent component of geographic variation in cancer burden while smoothing over year-to-year fluctuations. Because all three years predate OxyContin’s introduction, the averaged measure cannot be affected by opioid-induced changes in health behavior.

Results. Table A.14 (column 6) shows that the dynamic pattern is preserved: the pre-period coefficient is essentially zero (0.009, SE = 0.907), the effect is negative and small during 1997–2000, and it rises to a peak of 3.182 in 2005–2007 ($p < .05$). The overall effects are attenuated by approximately 20% relative to the baseline, consistent with the averaged measure introducing noise: the earlier cancer rates (1980, 1985) are less predictive of OxyContin marketing intensity than the 1995 rate, which directly determined Purdue Pharma’s initial targeting decisions. The 2008–2014 effect (2.430) loses statistical significance, suggesting that the 1995 rate provides a sharper instrument for the specific opioid exposure channel. The preservation of the qualitative pattern with a different instrument measure supports the robustness of the core finding.

Check 9: Excluding the South Census Region

Referee concern. The prescription opioid epidemic was geographically concentrated in Appalachian and Southern states, where OxyContin marketing was most aggressive and where pre-existing socioeconomic conditions — including high poverty rates, low educational attainment, and limited access to health services — may independently predict teen birth trends. If the results are driven entirely by these historically disadvantaged Southern communities, the findings may reflect region-specific dynamics rather than a generalizable relationship between opioid exposure and teen fertility.

Test. We exclude all commuting zones located in the South Census region, reducing the sample from 621 to 345 CZs. This removes the region where prescription opioid exposure was most intense and where many of the communities most affected by the crisis are located (e.g., West Virginia, Kentucky, Tennessee, and other Appalachian states).

Results. Table [A.14](#) (column 7) shows that the effects are attenuated and lose statistical significance when the South is excluded. The peak effect in 2005–2007 falls from 3.964 to 2.437 ($SE = 1.978$), and the 2008–2014 effect declines from 3.371 to 1.846 ($SE = 2.005$). However, the dynamic *pattern* is preserved: coefficients are near zero in the pre-period (0.167), negative in 1997–2000 (-1.358 , significant at 5%), and peak during 2005–2007 before declining. The attenuation reflects two factors: (i) the South experienced the earliest and most intense prescription opioid exposure, so removing it removes the strongest treatment signal; and (ii) the substantially smaller sample (345 vs. 621 CZs) reduces statistical power. The preservation of the dynamic pattern even in the non-South sample suggests that the mechanism linking opioid exposure to teen births is not uniquely Southern but rather operates wherever opioid exposure was sufficiently intense.

Check 10: State-Level Clustering of Standard Errors

Referee concern. The baseline specification clusters standard errors at the commuting zone level (621 clusters), which accounts for serial correlation within CZs over time and cross-sectional correlation within CZs across individuals. However, if opioid-related shocks are correlated across CZs

within the same state — due to common state-level policies, shared media markets, or correlated economic conditions — then CZ-level clustering may understate the true standard errors, leading to over-rejection of the null hypothesis.

Test. We re-estimate the baseline specification clustering standard errors at the state level (49 clusters) rather than the CZ level (621 clusters). This allows for arbitrary within-state correlation in the error terms across both CZs and time periods.

Results. Table A.14 (column 8) reports the results. Point estimates are mechanically identical to the baseline, as only the variance estimation changes. Standard errors increase by approximately 20–30% under state-level clustering — for example, the 2005–2007 standard error rises from 1.500 to 1.950, and the 2008–2014 standard error increases from 1.514 to 1.937. Despite these larger standard errors, the 2005–2007 peak effect of 3.964 remains significant at the 5% level ($t = 2.03$), and the 2008–2014 effect of 3.371 is borderline significant at the 10% level ($t = 1.74$). The qualitative conclusions are unchanged under this more conservative inference approach.

Check 11: Commuting Zone–Specific Linear Time Trends

Referee concern. Different commuting zones may have been on different secular trajectory paths for teen birth rates, driven by long-run demographic, economic, or cultural changes unrelated to the opioid crisis. If areas with high cancer mortality were already on steeper downward (or upward) trends in teen births before 1996, the estimated post-OxyContin effects could reflect the continuation of these pre-existing trends rather than a causal effect of opioid exposure. This concern is particularly relevant given the dramatic nationwide decline in teen birth rates since the early 1990s, which may have proceeded at different rates across communities.

Test. We augment the baseline specification with CZ-specific linear time trends, allowing each of the 621 commuting zones to have its own linear trend in teen birth rates over the sample period. This is a demanding test: it absorbs any variation in teen births that follows a linear path within each CZ, requiring the opioid effect to be identified from deviations from each CZ’s own trend line.

Results. Table A.14 (column 9) shows that the results are virtually identical to the baseline. The 2005–2007 peak effect is 3.961 (baseline: 3.964), the 2008–2014 effect is 3.368 (baseline: 3.371), and all standard errors change by less than 0.01. The CZ-specific linear trends are absorbed without meaningfully altering the estimated treatment effects. This indicates that the baseline results are not driven by differential linear trends across commuting zones. The opioid effect represents a *deviation* from each community’s own trend, consistent with the discrete shock interpretation: OxyContin’s introduction in 1996 generated a break in the relationship between area-level cancer mortality and teen birth trajectories that cannot be explained by pre-existing linear dynamics.

Check 12: Baseline Characteristics Interacted with Year Trends

Referee concern. Areas with high cancer mortality in 1995 tend to share a constellation of socioeconomic characteristics: they are more rural, more manufacturing-dependent, have lower incomes per capita, and have higher smoking prevalence. Each of these baseline characteristics could independently predict differential trends in teen birth rates through channels unrelated to opioids. For example, the decline of manufacturing employment may have reduced economic opportunities for young people, encouraging earlier family formation; rural areas may have experienced different trends in access to contraception and reproductive health services; and areas with high smoking prevalence may have experienced correlated health behavior changes. If these baseline characteristics drive differential trends in teen births that happen to coincide with the post-OxyContin period, the estimated opioid effect would be spurious.

Test. We add four sets of baseline characteristic $\times$ year interactions as controls: (i) manufacturing employment share in 1994 $\times$ year, (ii) log income per capita in 1994 $\times$ year, (iii) a rural indicator based on 1993 urban-rural classification $\times$ year, and (iv) county-level smoking prevalence $\times$ year. These controls allow each baseline characteristic to have a flexibly evolving relationship with teen birth rates over the sample period, absorbing any trends in teen births that are linearly associated with these area-level features.

Results. Table A.14 (column 10) reports the estimates. The dynamic pattern is preserved: the

pre-period coefficient is mechanically zero (the reference period), the 1997–2000 effect is -0.680 ($SE = 0.394$), and the peak effect in 2005–2007 is 3.405 ($p < .05$). The effects are modestly attenuated relative to the baseline — the 2005–2007 peak declines from 3.964 to 3.405 (a 14% reduction) and the 2008–2014 effect declines from 3.371 to 2.816 (a 16% reduction) — which is expected: some variation in the cancer mortality instrument is correlated with these baseline characteristics, so controlling for their trends absorbs part of the opioid signal. The fact that substantial, statistically significant effects survive after simultaneously controlling for manufacturing decline trends, income trends, rurality trends, and smoking prevalence trends provides strong evidence that the results are driven by the opioid exposure channel rather than by pre-existing socioeconomic trajectories. The sample is slightly smaller ($N = 11,343$; 614 CZs) due to missing values in some baseline characteristics.

Table A.14: Robustness of Grouped Period Estimates to Alternative Specifications

	(1) Baseline	(2) 1980 Cancer Placebo	(3) Drop Top/ Bottom 5%	(4) Pop.- Weighted	(5) Great Rec. Control	(6) Avg. Cancer 1980–1990	(7) Drop South	(8) State Clustering	(9) CZ Linear Trends	(10) Baseline Char. $\times$ Year
Pre (1990–1995)	−0.291 (0.813)	0.137 (0.932)	−0.483 (0.922)	−0.302 (0.855)	−0.317 (0.814)	0.009 (0.907)	0.167 (0.987)	−0.291 (0.631)	−0.291 (0.809)	0.000 (0.000)
1997–2000	−0.619 (0.392)	−0.794 (0.414)	−0.538 (0.432)	−0.674 (0.471)	−0.622 (0.392)	−0.748 (0.411)	−1.358** (0.598)	−0.619 (0.344)	−0.619 (0.393)	−0.680 (0.394)
2001–2004	2.400* (1.232)	1.605 (1.249)	2.632* (1.320)	−0.329 (0.964)	2.218* (1.206)	1.954 (1.266)	1.100 (1.716)	2.400 (1.650)	2.397* (1.235)	1.961 (1.250)
2005–2007	3.964*** (1.500)	2.624 (1.474)	4.216*** (1.610)	0.280 (1.409)	3.735** (1.464)	3.182** (1.503)	2.437 (1.978)	3.964** (1.950)	3.961*** (1.500)	3.405** (1.513)
2008–2014	3.371** (1.514)	1.840 (1.472)	3.697** (1.634)	−0.098 (1.312)	3.205** (1.465)	2.430 (1.504)	1.846 (2.005)	3.371* (1.937)	3.368** (1.513)	2.816* (1.523)
2015–2020	1.477 (1.528)	0.600 (1.476)	2.108 (1.645)	−0.519 (1.189)	1.299 (1.474)	0.934 (1.507)	−0.648 (2.097)	1.477 (1.750)	1.476 (1.527)	1.194 (1.527)
Observations	13,975	13,975	12,199	13,864	13,975	13,975	7,814	13,975	13,975	11,343
Clusters	621	621	544	621	621	621	345	49	621	614

Notes: Each column reports estimates from a separate regression of the teen birth rate (per 1,000 women aged 15–19) on the interaction of the treatment variable with grouped period indicators. The reference period is 1996. All specifications include commuting zone and state $\times$ year fixed effects unless otherwise noted. Standard errors clustered at the commuting zone level are in parentheses, except column (8) which clusters at the state level. Column (1): baseline specification using 1995 cancer mortality. Column (2): 1980 cancer mortality as placebo treatment. Column (3): drops CZs in the top and bottom 5% of the population distribution. Column (4): weighted by CZ population. Column (5): adds Great Recession exposure $\times$ post-2007 control. Column (6): uses the average of 1980, 1985, and 1990 cancer mortality rates (standardized). Column (7): excludes the South Census region. Column (8): clusters standard errors at the state level (49 clusters). Column (9): adds CZ-specific linear time trends. Column (10): adds baseline characteristics (manufacturing share, log income per capita, rural indicator, smoking prevalence) interacted with year.

* $p < .10$, ** $p < .05$, *** $p < .01$.

Table A.15: Pre-Period Placebo: 1980 Cancer Mortality and Teen Births, 1980–1995

	Teen Birth Rate (per 1,000)
1980 Cancer $\times$ Pre (1980–1984)	0.715 (0.491)
1980 Cancer $\times$ Post 1 (1986–1990)	–3.016 (1.832)
1980 Cancer $\times$ Post 2 (1991–1995)	–2.769 (1.429)
Reference period	1985
Observations	3,952
Commuting zones	617
CZ fixed effects	Yes
State $\times$ year FE	Yes

Notes: This table reports estimates from a regression of the teen birth rate on the interaction of 1980 age-adjusted cancer mortality with grouped period indicators, using only data from 1980–1995 (before OxyContin’s introduction). The reference year is 1985, serving as a placebo “treatment” date. The specification includes commuting zone and state $\times$ year fixed effects. Standard errors clustered at the commuting zone level are in parentheses. No positive effects emerge, confirming that the cancer mortality instrument does not predict teen birth dynamics in the pre-OxyContin era.

* $p < .10$, ** $p < .05$, *** $p < .01$.

Table A.16: NCHS Data Only: Teen Birth Estimates Using 2003–2020 Sample

	Teen Birth Rate (per 1,000)
Cancer ₁₉₉₅ × 2004–2007	0.663*** (0.168)
Cancer ₁₉₉₅ × 2008–2014	0.354 (0.399)
Cancer ₁₉₉₅ × 2015–2020	−1.570*** (0.596)
Reference year	2003
Observations	11,178
Commuting zones	621
CZ fixed effects	Yes
State × year FE	Yes

Notes: This table reports estimates using only the NCHS Bayesian-imputed county-level birth data (2003–2020), with 2003 as the reference year. The specification includes commuting zone and state × year fixed effects. Standard errors clustered at the commuting zone level are in parentheses. The significant positive effect in 2004–2007 confirms that the main finding is not an artifact of the splice between NBER Natality and NCHS data sources. Magnitudes are smaller than the baseline because the reference year (2003) already falls within the “positive effect” window, compressing the post-treatment contrast.

* $p < .10$, ** $p < .05$, *** $p < .01$.